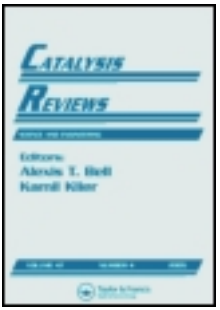
This article was downloaded by: [Tulane University]
On: 05 October 2013, At: 09:26
Publisher: Taylor & Francis
Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Catalysis Reviews: Science and Engineering
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/lctr20

Methane-Steam Reforming

James P. Van Hook [a] [b]

[a] Research and Development Center, Pullman Kellogg Houston, Texas, 77084

[b] Foster Wheeler Energy Corp., 110 South Orange Ave., Livingston, New Jersey, 07039

Published online: 06 Dec 2006.

To cite this article: James P. Van Hook (1980) Methane-Steam Reforming, Catalysis Reviews: Science and Engineering, 21:1, 1-51

To link to this article: http://dx.doi.org/10.1080/03602458008068059

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any

form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at http://www.tandfonline.com/page/terms-and-conditions

Downloaded by [Tulane University] at 09:26 05 October 2013

CATAL. REV.—SCI. ENG., 21(1), 1-51 (1980)

Methane-Steam Reforming

JAMES P. VAN HOOK*
Research and Development Center
Pullman Kellogg
Houston, Texas 77084

Downloaded by [Tulane University] at 09:26 05 October 2013

I. INTRODUCTION 2

II. HISTORICAL 3

III. EARLY KINETICS AND CATALYST FORMULATIONS (1933-1952) 5

IV. GENERAL APPROACH TO KINETICS OF METHANE-STEAM REFORMING, WATER-GAS SHIFT, AND CARBON-FORMING TENDENCY 10
A. Water-Gas Shift during Methane-Steam Reforming 12
B. Carbon-Forming Tendency during Methane-Steam Reforming 16

V. KINETICS OF CATALYZED METHANE-STEAM REFORMING 24
A. Overall Kinetic Model for Steam Reforming and Water-Gas Shift Equilibration 26
B. Summary of Temperature Effect in Methane-Steam Reforming 46

VI. CONCLUDING REMARKS 48

REFERENCES 48

*Current address: Foster Wheeler Energy Corp., 110 South Orange Ave., Livingston, New Jersey 07039.

Copyright © 1980 by Marcel Dekker, Inc.

I. INTRODUCTION

For more than 40 years the catalyzed hydrocarbon-steam reforming reaction has been of commercial value. The first units commissioned in the 1930s were run on light hydrocarbon feedstock at atmospheric pressure [1]. It took until 1962 for a 15-atm steam reformer to come on-stream in the United Kingdom using liquid hydrocarbon feedstock [2-4].

On paper, the reaction simply produces H_2, CO, CO_2, and CH_4 in various proportions depending on the reaction conditions dictated by the end use of the gas mixture desired. The pertinent equations describing the system are:

$$C_xH_y + xH_2O = xCO + (y/2 + x)H_2$$

$$CO + H_2O = CO_2 + H_2$$

$$CO + 3H_2 = CH_4 + H_2O$$

or the four basic equilibria for any hydrocarbon system:

$$C_xH_y = xC + y/2H_2$$

$$CH_4 = C + 2H_2$$

$$2CO = CO_2 + C$$

$$CO + H_2 = C + H_2O$$

Ammonia, methanol, Fischer-Tropsch synthesis gas, hydrogen itself, metal ore reducing gas, Town gas, and, more recently, substitute (synthetic) natural gas are the worldwide uses of the catalytic steam reforming reaction.

The product gas produced in the primary reforming process depends on the operating temperatures, pressure, and steam and hydrocarbon feed rates, and with sufficient catalyst present, closely approaches the thermodynamic equilibria inherent in the system. Thus, for ammonia synthesis, the final gas should have a H/N atom ratio of about three, obtained by the most economically designed process involving pressure, temperature, steam-to-hydrocarbon feed ratio throughput, heat flux and furnace design, materials of construction, heat recovery, gas compression, mass transfer, pressure drop allowances, and last but not least, catalysis with internal pore diffusion

Downloaded by [Tulane University] at 09:26 05 October 2013

limitations. Heterogeneous catalysis, equilibria, and diffusion using nickel catalysts are specific to this review. Large pressure (0.01 to 50 atm), temperature (500 to 1700°F) and intrinsic catalyst activity (200,000-fold) ranges are covered herein. Various adsorption studies, catalyst poisoning, and regeneration work and decomposition studies in the absence of steam are covered elsewhere in a review by Rostrup-Nielson [5].

Two rather extreme conditions in the steam reforming process are: (1) reducing gas production which involves primary steam reforming at high temperature (1800°F and higher), low pressure (about 50 psi to minimize methane concentration), and low steam-to-hydrocarbon feed ratio (minimize unreacted steam and/or CO_2, which would have to be condensed or absorbed before the gas is injected into the ore reduction reactor); and (2) substitute natural gas (SNG) production, which requires steam reforming at the highest possible pressure (methane thermodynamically favorable), the lowest steam-to-carbon ratio (moles of steam per carbon atom in the feedstock) possible without carbon formation (again favorable methane formation), and the lowest possible operating temperature (kinetics permitting). Certainly, one would like to react carbon and steam at the lowest temperature to produce essentially CO_2 and CH_4.

As simple as the reaction and equilibria might appear, there are a large number of reactions occurring simultaneously, such as hydrocarbon cracking, polymerization, steam reaction with surface carbon-hydrogen residues, methanation, hydrocracking (hydrogenolysis), dehydrogenation, carbon monoxide disproportionation, water-gas shift, and even dehydrocyclization reactions, all apparently associated with a steady state of carbon-hydrogen-oxygen surface residues on the catalyst. Since several reactions are "carbon" producers and others are "carbon" removing reactions, considerable attention must be given to the catalyst and operating variables in order to maintain an acceptable level of steady-state "carbon" on the catalyst. Many types of primarily nickel catalysts containing various support materials and promoters have been suggested in order to achieve the highest physical strength, highest catalyst activity for converting feed, and lowest tendency to produce carbon (highest tendency to remove it).

II. HISTORICAL

In 1780 Fontana first observed the production of a combustible gas when steam was passed over red hot coke. The product gas from such a reaction was first analyzed and reported 22 years later [6]. By the

Downloaded by [Tulane University] at 09:26 05 October 2013

19th century, "illuminating gas" was used, having been produced by the simple destructive distillation of coal. Hydrocarbon interaction with catalytic metals was observed in 1817 by Sir Humphrey Davy [7] during the development of the mine safety lamp. The production of hydrogen when hydrocarbons and steam are passed over calcium oxide was described in 1868 [8], and a two-step conversion process using nickel was claimed in 1889 by Mond and Langer [9]. Solid fuels were gasified with air and steam in the latter 19th century, and with steam in a cyclic operation in order to supply the large heat required of the steam-carbon reaction [10]. Vignon [11] apparently was the first to observe alkaline earth catalysis of the steam-carbon reaction.

Patents appeared early in the area of hydrogen generation by water-gas shifting using transition metals [12, 13], and in the area of steam reforming methane over a supported nickel catalyst [14] following Mond and Langer's 1889 two-step process involving deposition of carbon from carbon monoxide or hydrocarbons followed by carbon removal with steam. A 1927 steam reforming patent [15] claimed low operating temperature by using catalysts of iron, cobalt, or nickel promoted with chromia, vanadia, alkali, or alkaline earths.

The very early work pertaining to the reactions of CO, CO_2, H_2, H_2O, CH_4, and carbon-on-nickel catalysts was summarized by Sabatier in 1922 [16]. Very frequent notation was made to the problems associated with carbon formation.

A state-of-the-art summary in the area of steam reforming to the year 1931 was covered by Marek and Hahn [17], in which period equilibrium considerations [18] and rudimentary studies in chemical kinetics began to emerge. Using nickel catalysts, theoretical values for equilibria in the methane-steam system were verified [19, 20], and the experimental demonstration of the effect of temperature and pressure on the $CO_2 + CH_4 = 2CO + 2H_2$ equilibrium was shown by Pichler in 1931 [21], who also demonstrated that reaction velocity was inversely proportional to the total operating pressure. This was probably the first indirect reference to pore diffusional limitations in the catalytic steam reforming reaction.

Fischer and Tropsch [22] screened a large number of catalysts for the steam-methane and CO_2-methane reactions at temperatures of 1600 to 1800°F, and found that supported nickel or cobalt rated as the best preparations. Some catalyst aging occurred, but stability resulted after a period of time.

Describing a cyclic steam or CO_2 process for converting methane, Hawk et al. [23] and Karzhavin [24], who used impregnated nickel catalysts with reaction heat being supplied by air-gas combustion,

Downloaded by [Tulane University] at 09:26 05 October 2013

stated that in steam reforming methane, complications such as the appearance of tarry substances and heavy members of the benzene series can arise. Serious difficulties with "carbon" formation were envisioned in these 1932-1936 reports in steam reforming heavier hydrocarbons. (Fairly recent work by Bhatta and Dixon [25] in steam reforming butane at 850°F, 30 atm., and a steam-to-carbon ratio as low as 1.0 showed severe catalyst deactivation and proved the existence of polynuclear aromatics on the catalyst. Even with a high degree of deactivation, the catalyst was restored to "approximately" the original activity by hydrogen treatment at 930°F for 16 hr.)

III. EARLY KINETICS AND CATALYST FORMULATIONS (1933-1952)

In 1933 Fujimoto [26] reported on methane-steam reforming and methane decomposition over a broad temperature range of 752 to 1832°F using reduced nickel foil and a 90% nickel-alumina catalyst at atmospheric pressure.

For steam reforming at a steam-to-carbon ratio of about 2.5, and presumably constant feed rate conditions, equilibrium methane conversions were achieved using the 90% Ni/alumina catalyst over the entire temperature range studied. Using the nickel foil, equilibrium methane conversions were obtained from 750 to 1050°F, identical to the results obtained with the nickel-alumina. However, increasing the temperature beyond 1100°F with the foil resulted in a serious loss in activity with only a few percent methane conversion being attained between 1300 and 1650° F. The rate then increased as temperature rose, until 70% methane conversion was achieved at 2000°F. The nickel foil had been seriously damaged with carbon by operating above 1100°F, whereas the 90% nickel-alumina withstood 250 hr of operation at 1292°F and maintained equilibrium conversion.

It is interesting to note that when nickel foil is used in steam reforming methane, the steam reforming activity declines at the temperature where methane decomposition, in the absence of steam, begins. In the low temperature region, 750 to 1060°F, steam reforming is fast compared to methane decomposition. Even at 1022° F, for instance, the steam-methane reaction was essentially at equilibrium, but in the absence of steam no significant methane conversion had yet been obtained. It was not until 1100° F, where the methane alone begins to decompose, that the nickel foil began to lose activity for steam reforming. This suggests that the nickel surface is reduced in activity

Downloaded by [Tulane University] at 09:26 05 October 2013

by small amounts of "carbon" when steam is not present. Steam rapidly removes this "carbon" even at low temperatures (<1100°F) and steam reforming proceeds. Above 1100° F, the steady-state amount of "carbon" increases substantially, resulting in serious deactivation of the nickel surface. When the temperature is further increased to about 1600°F, where the methane decomposition reaction approaches equilibrium (in the absence of steam), the steam "methane" reaction rate now increases and continues to do so as the temperature is further increased. This high-temperature (>1600°F) steam-reforming region suggests that, in reality, the reaction is the steam-carbon reaction, with carbon produced by methane decomposition. At about 1800° F the methane decomposition reaction is equilibrated with respect to bulk carbon, and steam is gasifying the carbon. An Arrhenius plot of these high-temperature "steam-reforming" data yields an apparent activation energy of 98,000 Btu/lb mole, reasonably close to the 100,000 to 145,000 values reported by many investigators of various carbon-steam reactions [27].

Assuming, as with the steam-reforming data, that the methane decomposition reaction can be approximated by reversible first-order (in methane) kinetics, Fujimoto's data (shown in Fig. 1) were subjected to Arrhenius plots in which the lowest temperature was arbitrarily chosen to have a contact time of 1 sec and all other temeratures were corrected for RT gas expansion. The Arrhenius plot of the methane decomposition data (thermodynamic equilibrium on the basis of carbon, as graphite, at unit activity) for the nickel foil yields an excellent straight line from 1300 to 1650° F with an apparent activation energy of 78,000 Btu/lb mole.

In the same manner, an Arrhenius plot of the methane decomposition data over the nickel-alumina catalyst, where decomposition had already commenced at 750° F, an activation energy of only 9,500 Btu/lb mole is apparent in the 750 to 1100°F range. An increase in the activation energy apparently takes place above 1200°F, but the data at 1382° F and higher are too close to thermodynamic equilibrium (>95% approach) to yield a proper estimate.

The behavior of these two reactions when a high activity (metal area per unit of catalyst volume) nickel-alumina is used (internal and/or bulk diffusion limitations apparent, E_{act} = 9500) relative to nickel foil (E_{act} = 78,000), suggests that limiting the reactions by introducing diffusional resistance substantially increases operability and catalyst life.

Downloaded by [Tulane University] at 09:26 05 October 2013

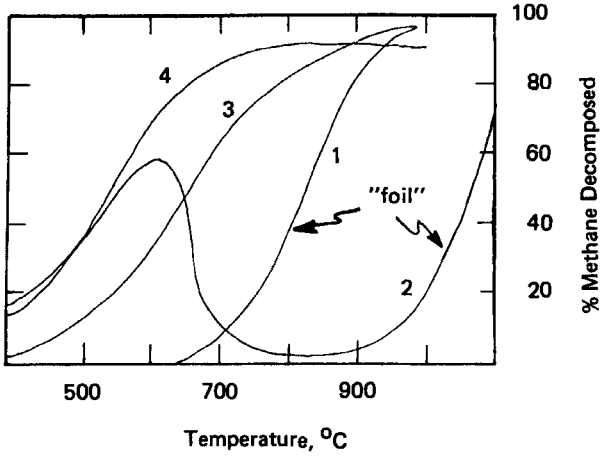
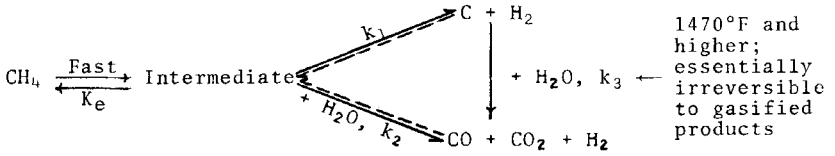


FIG. 1. Methane decomposition and steam reforming (after Fujimoto [26]). Curve 1: Reduced nickel, CH_4 decomposition. Curve 2: Reduced nickel, CH_4 steam reforming. Curve 3: 90% Ni/alumina, CH_4 decomposition. Curve 4: 90% Ni/alumina, CH_4 steam reforming.

A reasonable mechanistic scheme, based on Fujimoto's data for nonporous nickel foil (A) and high activity porous nickel-alumina catalyst (B) might be:

(A) Methane decomposition/steam reforming on nonporous foil:



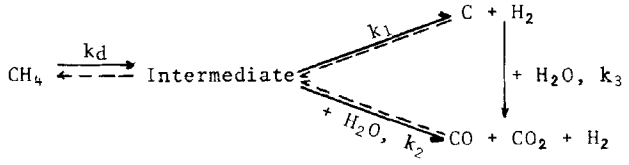
where intermediate concentration = $K_e P_{CH_4}/P_{H_2}{}^x$ and $k_2 > k_1 > k_3$, at $\leq 1100^\circ F$, and

E_{act} for k_3 = 98,000 Btu/lb mole

E_{act} for k_1 = 78,000 Btu/lb mole

Downloaded by [Tulane University] at 09:26 05 October 2013

(B) Methane/steam reaction on porous catalyst with diffusion limitations:



where k_d is the internal pore diffusion rate constant and can be slowed down relative to the inherent gasification rate constant by using porous catalysts with a high internal activity per unit of geometric surface area.

In 1946, Gordon [28] studied natural gas (10.5% ethane and propane) uncatalyzed steam reforming at 2230 to 2760°F and steam-to-gas ratios of 1.5 to 5. High conversions were attained, but considerable amounts of carbon were formed (up to 53% of catalyst bed).

System equilibria for CH_4/steam and CH_4/CO_2 and operational areas where carbon would/would not be expected to form were put forth by Reitmeier et al. in 1948 [29].

Studies were performed on the CH_4/steam and CH_4/CO_2 reforming reaction using finely divided copper-on-silica in a fluid bed, and early mechanism suggestions were developed in 1949 by Lewis et al. [30]. Besides studying the steam and CO_2 reforming of methane on reduced copper, they also replaced the feed oxidant (CO_2 or steam) with copper oxide in a circulating fluid bed. In both sets of experiments they found that the results were essentially the same, showing that the reduction of copper oxide in the 1400 to 1600°F range is fast compared to the reaction of product CO_2/steam with methane. They indicated that the kinetics show first-order dependence in both steam (or CO_2) and methane. The poor performance of copper as a catalyst was shown by the low conversion rates of 0.01 to 0.03 g moles CH_4 converted/g cat/hr/atm CH_4 at 1500° F. Noteworthy in this early work for both CO_2 and steam reforming of methane is the high activation energy of 86,000 Btu/lb mole for finely divided catalyst and the appearance of inhibition terms (CO_2, H_2O, H_2) in the rate expression.

Ipatieff et al. [31] screened various catalysts for methane-steam reforming in the 900 to 1500°F range, 1 and 18 atm pressure, and steam-to-carbon ratios of 5, 11, and 30. Catalysts based on copper, cobalt, iron, and several combinations of these metals gave poor activity. Nickel-on-kieselguhr catalysts showed high activity. Life

Downloaded by [Tulane University] at 09:26 05 October 2013

tests indicated that a 7.4% copper, 65% nickel-on-kieselguhr was the best and showed the beneficial effect of copper promotion on the life of the catalyst. Operating pressure, even at a steam-to-carbon ratio of 32, severely affected catalyst life, especially the nonpromoted nickel catalyst.

Early attempts to develop sulfur-resistant steam-reforming catalysts were reported by Sebastian and Riesz in 1951 [32]. Supported nickel, cobalt, iron, molybdenum, chromium, and combinations of these metals were tested. In contrast to the low activity of the copper catalyst (0.02 at 1500° F) for steam-reforming methane in the absence of sulfur reported by Lewis [30], Sebastian and Riesz, using their best sulfided nickel-on-silica (Ni/Si = 1/3 atom ratio), achieved the high rate of propane conversion in the presence of 2500 ppm H_2S in propane of 0.2 g moles C_3 converted/g cat/hr/atm propane at their lowest temperature studied, 1200° F.

Shortly after the sulfur-resistant catalyst studies, Reisz et al. [33], in screening 33 nickel-impregnated supports in a standard test, showed that magnesia and alumina were uniformly superior supports for nickel and that higher nickel content is necessary for maximum activity as the porosity of the support is increased (3.5% Ni for 24% porosity and 6 to 9% Ni for 52% porosity).

In 1952 Arnold et al. [34] reported on the activities of nine commercial catalysts for steam-reforming methane at atmospheric pressure and a steam-to-carbon ratio of 2.5 moles/mole. Space velocity and temperature were varied using mostly 16/20 mesh catalyst.

In general, it was shown that catalysts with higher nickel content had higher activity. Catalyst particle size was mentioned as being very important in the activity test. Reasonable reversible first-order (in methane) kinetic plots, ln (100 - % approach to steam methane equilibrium) vs contact time, are obtained with data from all nine catalysts where sufficient data are available. A set of data at various space velocities and 1300 and 1500°F using 16/20 mesh 6% nickel in alumina/silica at atmospheric pressure yield rate constants of 9.3 and 52.3 sec^{-1} at the respective temperatures (contact time on the basis of solid catalyst volume and gas at temperature). These values suggest an activation energy of at least 60,000 Btu/lb mole.

Semipilot plant (2.75 in. i.d. tube) data presented, using 7/16 X 7/16 in. extruded catalyst (24% Ni in $Ca/SiO_2/Al_2O_3$) at various space velocities and 1400 to 1550°F, likewise permitted calculation of rate constants at 4.5 and 7.8 sec^{-1}, respectively, for an indicated activation energy of 26,000 Btu/lb mole. This is probably the earliest published data showing a change of activation energy with a change in

Downloaded by [Tulane University] at 09:26 05 October 2013

catalyst particle size, and occurred some 20 years after Pichler's work [21] suggested a lowering of E_{act} as the total pressure is increased.

Arnold's attempts to steam-reform methane at 2100°F resulted in rapid plugging of the reactor with carbon. Hydrogen/steam aging tests (H_2/steam of 0.4 and 40 M) showed little, if any, damage to the activity of the nickel catalysts at 1180°F for 30 days. All catalysts were very seriously damaged at 1950°F.

IV. GENERAL APPROACH TO KINETICS OF METHANE-STEAM REFORMING, WATER-GAS SHIFT, AND CARBON-FORMING TENDENCY

The first extensive study of the kinetics of the steam-methane reaction was published by Akers and Camp [35] in 1955. Their work was performed at atmospheric pressure using 1/8 in. commercial nickel-on-kieselguhr catalyst in an integral reactor designed to minimize heat transfer problems associated with this highly endothermic reaction. Their data included a complete analysis of the product gases. Most of the study was performed at 1180°F by changing the contact time with a constant catalyst bed size at steam-to-carbon ratios of approximately 9.9, 4.7, 3.3, and 2.4. The kinetics showed an excellent fit to first order in methane at all steam-to-carbon ratios studied, and at a 10 to 85% approach to the steam-methane thermodynamic equilibrium. There was evidence in this work of pore diffusion limitations, as expected for a moderately active catalyst of 1/8 in. size. No experiments were performed using larger or crushed catalyst. An apparent activation energy of 15,800 Btu/lb mole was obtained with evidence of increasing E_{act} at the lowest temperature studied (637° F).

Bulk mass transfer of reactants to the catalyst external surface showed very little, if any, limitation, as evidenced by the constancy of the rate constant (0.95 g moles methane converted/g cat/hr/atm methane at 1180° F) at all S/C ratios and over a range of superficial velocities of 0.2 to 3.5 ft/sec. This rate is considerably lower than the bulk mass transfer rate (3 to 20 g moles/g cat/hr/atm CH_4 at 0.2 to 4.0 ft/sec and catalyst geometric area of 14.8 cm^2/cm^3 bulk volume) calculated using the low Reynolds number correlation of Petrovic and Thodos [36]. In all cases then, bulk mass transfer of reactants to the catalyst pellet surface is expected to be considerably faster than the measured rate of methane disappearance, as shown in Fig. 2.

Downloaded by [Tulane University] at 09:26 05 October 2013

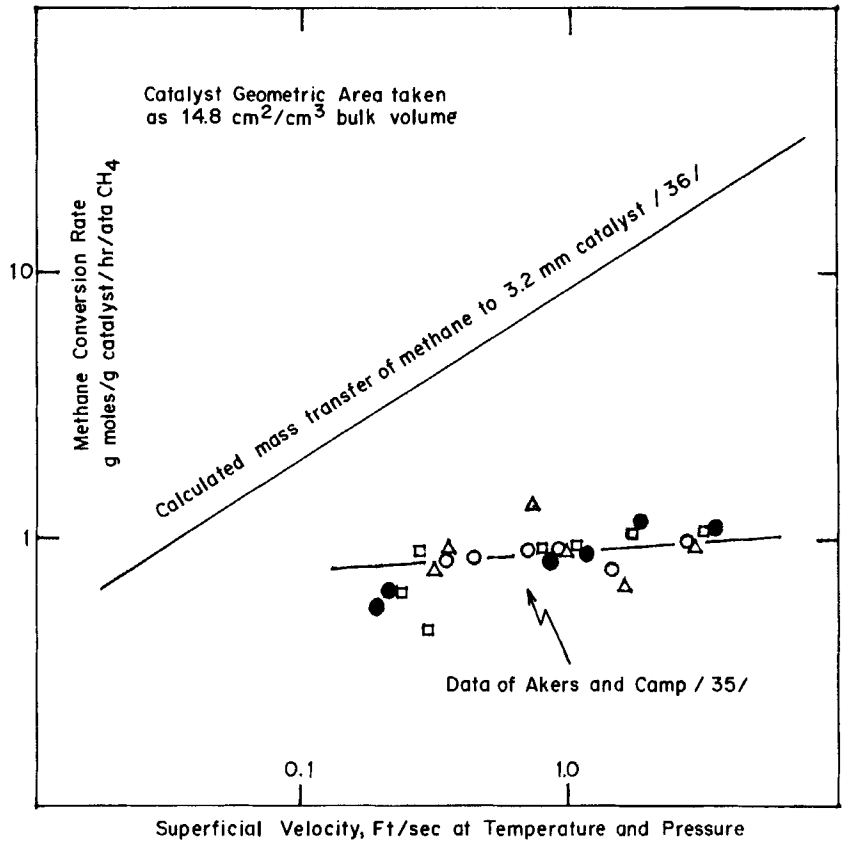


FIG. 2. Bulk mass transfer to external catalyst surface in 1180°F atmospheric pressure steam reforming of methane.

The authors presented their data as irreversible first-order (in methane) kinetics. An excellent fit to reversible first order is, however, also realized. (There are only a few data points higher than 60% approach to equilibrium methane conversion, where back reaction would be expected to contribute to a nonlinear plot.) At 1180°F, atmospheric pressure, and all steam-to-carbon ratios studied, their rate expression is $dCH_4/dt = kP_{CH_4}$, where k = 0.95 g moles CH_4 converted/g cat/hr/atm CH_4(39.4 sec^{-1}) with an activation energy of 15,800 Btu/lb mole.

Downloaded by [Tulane University] at 09:26 05 October 2013

In their analysis of the data, Akers and Camp were primarily concerned with the primary products of the methane-steam reaction and the approach to equilibrium for the three pertinent reactions that must be considered:

$$2CO \rightleftharpoons C + CO_2, \quad \Delta H = -74,600 \text{ Btu} \tag{1}$$

$$CH_4 \rightleftharpoons C + 2H_2, \quad \Delta H = +38,100 \text{ Btu} \tag{2}$$

$$CO + H_2 \rightleftharpoons C + H_2O, \quad \Delta H = -54,000 \text{ Btu} \tag{3}$$

and combinations of these such as:

(A) $CH_4 + H_2O \rightleftharpoons CO + 3H_2$, Steam-methane equilibrium (2 and 3)

(B) $CO + H_2O \rightleftharpoons CO_2 + H_2$, Water-gas equilibrium (1 and 3)

(C) $CO_2 + CH_2 \rightleftharpoons 2CO + 2H_2$, CO_2-methane equilibrium (1 and 2)

The authors concluded that both CO and CO_2 were primary products of the methane-steam reforming reaction. Only at the shortest contact times was the water-gas shift reaction strongly to the carbon monoxide rich side of equilibrium and appeared to be approximately the same for all S/C ratios studied. This is shown in Fig. 3, where ΔT_{wgs} (temperature deviation from shift equilibrium) is plotted as a function of methane conversion (or approach to equilibrium methane conversion, since methane conversions at equilibrium would all be high, i.e., 100, 95, 90, and 80% for S/C of 9.9, 4.7, 3.3, and 2.4, respectively). For Akers and Camp data, the water-gas shift reaction was essentially at thermodynamic equilibrium at 50% or greater methane conversion.

A. Water-Gas Shift during Methane-Steam Reforming

An effective presentation of Akers and Camp data is given in Fig. 4, where percent approach to shift equilibrium (CO_2/CO_2 at wgs equilibrium) and percent of CO that is present, at the measured methane conversion, which would have to be steam shifted in order to be in water-gas shift equilibrium, is plotted against methane conversion. This could be an extremely effective tool if a generalized correlation, independent of catalyst, temperature, and S/C ratio, could be developed, since only the methane conversion would need to

Downloaded by [Tulane University] at 09:26 05 October 2013

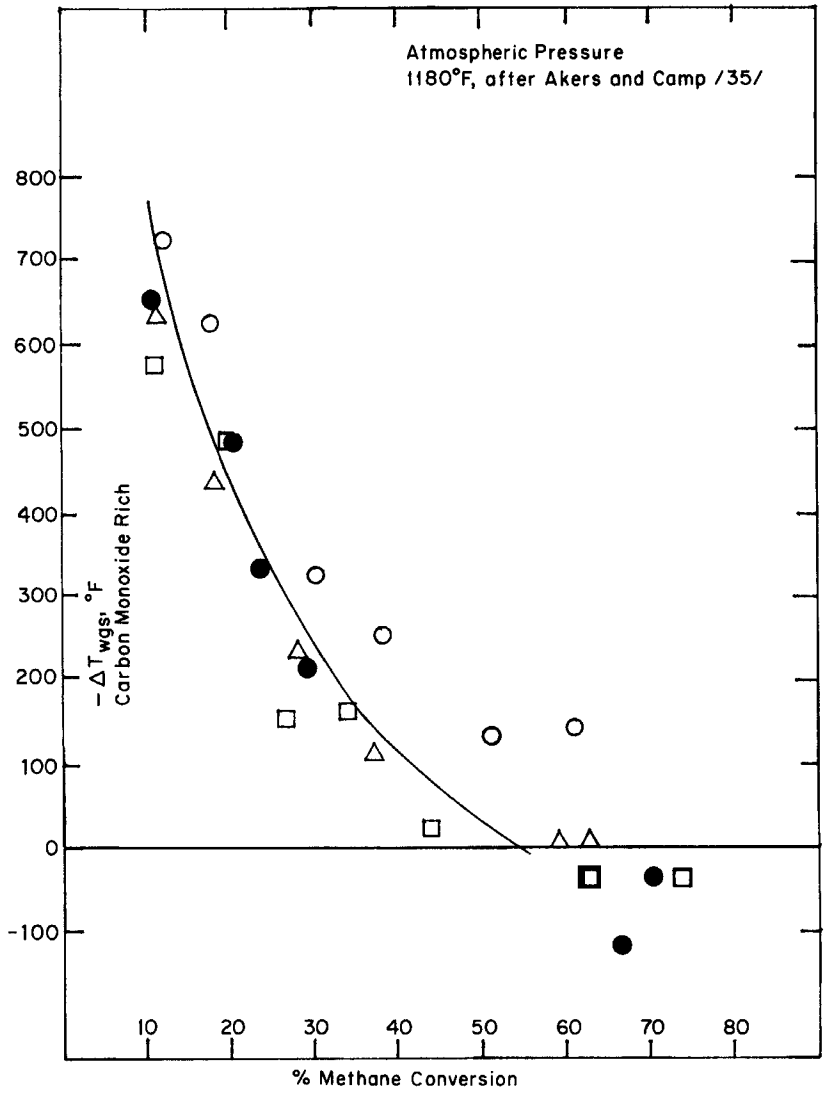


FIG. 3. Water gas shift reaction in methane steam reforming. O 100 Series S/C = 9.9 ± 0.4; Δ 200 Series S/C = 4.7 ± 0.5; □ 300 Series S/C = 3.3 ± 0.1; ● 400 Series S/C = 2.4 ± 0.3.

Downloaded by [Tulane University] at 09:26 05 October 2013

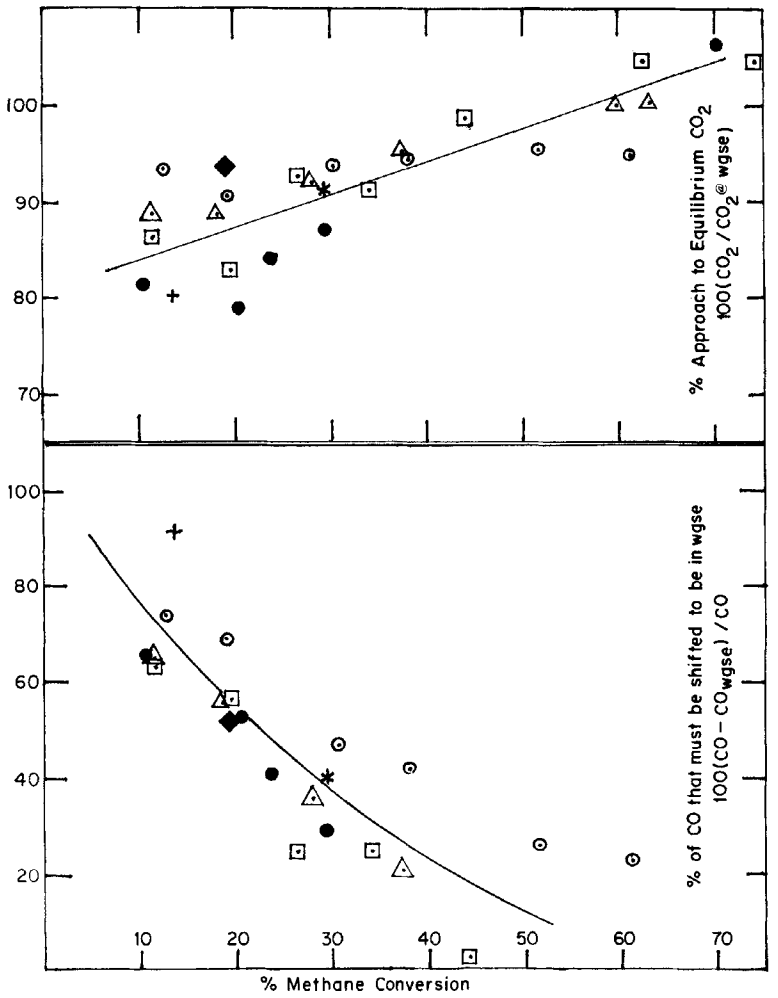


FIG. 4. Percent approach to shift equilibrium in methane steam reforming. See Fig. 3 legend. Also: + 749°F, S/C = 2.8; ◆ 842°F, S/C = 2.6; * 943°F, S/C = 2.8.

be known in order to determine the complete gas composition and danger of forming carbon. An interesting observation of Fig. 4 data is that the three low temperature data points (943, 842, and 749°F) fall on a simple methane conversion correlation line rather than approach to equilibrium, even though conversions of less than 30%

Downloaded by [Tulane University] at 09:26 05 October 2013

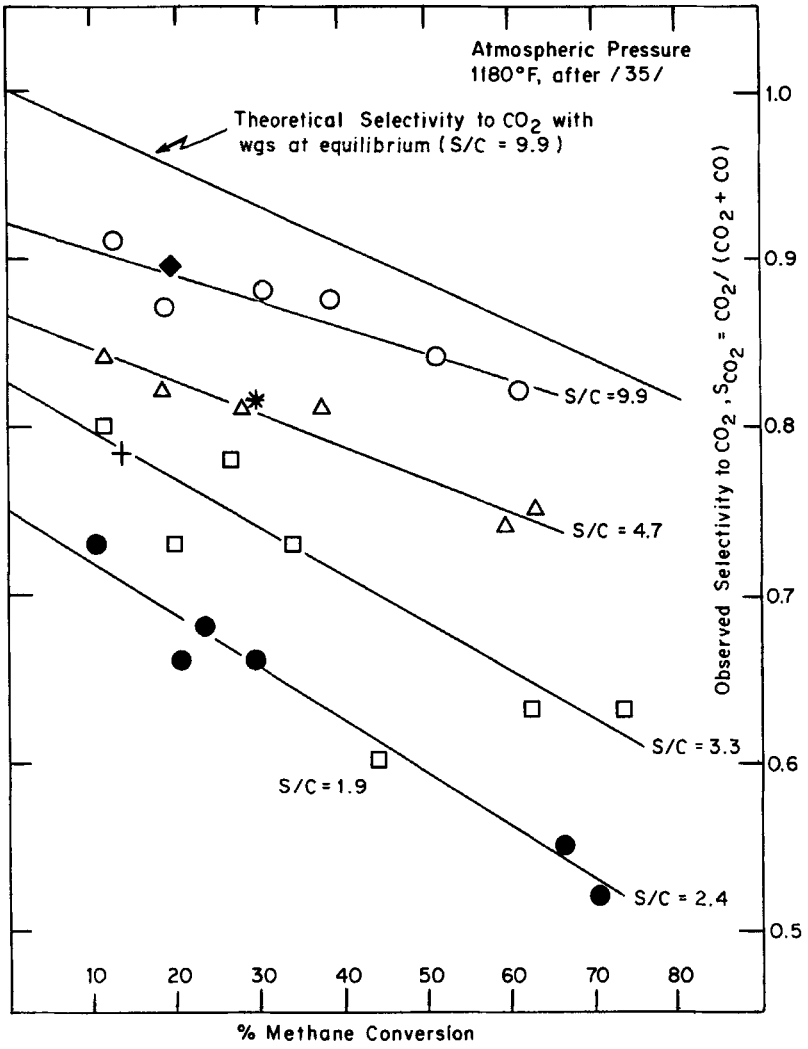


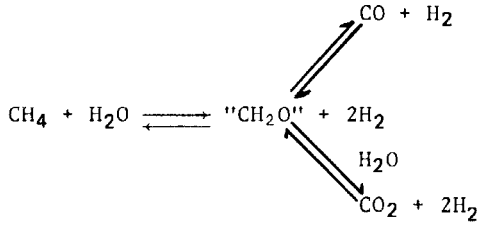
FIG. 5. Selectivity of steam reforming methane to carbon dioxide.

closely approach equilibrium methane conversion at these low temperatures.

Since the indicated selectivity to carbon dioxide (CO_2/conversion) at zero methane conversion, shown in Fig. 5, is directly proportional to the mole fraction of steam in the reacting mixture (it should approach 1.0 at all S/C ratios if wgs is in equilibrium), this is very

Downloaded by [Tulane University] at 09:26 05 October 2013

tempting evidence to suggest that CO and CO_2 are indeed both primary products, the surface C-H-O residue formed by dissociative chemisorption of methane and steam decomposed directly to CO and H_2, and also in a parallel route reacts with steam to produce CO_2 and H_2. The net balance of these two routes is simply the approach to water-gas shift equilibrium. As the steam pressure (mole fraction) is increased, the route to CO_2 increases in rate and the equilibrium position shifts more toward CO_2, resulting in the same approach to water-gas shift equilibrium.



B. Carbon-Forming Tendency during Methane-Steam Reforming

In all cases in the 1180°F operation of Akers and Camp [35] there was no tendency toward bulk carbon formation via Reactions (1) and (3), even with the water-gas shift not in equilibrium.

$$2CO = C + CO_2, \quad \Delta H = -74,600 \text{ Btu} \tag{1}$$

$$CO + H_2 = C + H_2O, \quad \Delta H = -54,000 \text{ Btu} \tag{3}$$

Obviously, at low methane conversion there is no thermodynamic protection from bulk carbon formation via a fast Reaction (2), since at low conversions the hydrogen pressure is low unless substantial amounts are added in the feed.

$$CH_4 = C + 2H_2, \quad \Delta H = +38,100 \text{ Btu} \tag{2}$$

We then have a balance between the rate of formation and removal of steady-state "carbon" and the thermodynamic driving force for producing and removing bulk carbon. Under all except four conditions (high methane conversion), bulk carbon formation by way of Reaction (2) was thermodynamically favorable. With Reactions (1), (2), and (3), and therefore naturally with Reactions (A), (B), and (C), at thermodynamic equilibrium at 1180°F and atmospheric pressure, there will be no bulk carbon formation until the steam-to-carbon feed ratio is lowered below approximately 1.4, which is the thermodynamic

Downloaded by [Tulane University] at 09:26 05 October 2013

Downloaded by [Tulane University] at 09:26 05 October 2013

minimum steam requirement. In all calculations of equilibria the activity of carbon was assumed to be equal to 1.0 for simplicity, since Reactions (1), (2), and (3) are all relative to carbon at some activity. Dent [37] has published thermodynamic data for these carbon-forming reactions on nickel. They have been summarized by Bridger and Chinchen [38] with additional data supplied by Rostrup-Nielson [5] in which the effect of temperature and pressure on these carbon-forming reactions is given. The activity of "carbon" relative to graphite, as determined by CO disproportionation, methane decomposition, methanation, and methane-steam reforming experiments over various nickel catalysts from 650 to 1600°F, is apparently as high as 50 at 650°F and decreases with increasing temperature to approximately unity (graphite) at 1500 to 1600°F. An effect of nickel crystallite size on the activity of carbon was also correlated with a clear indication of carbon whisker formation with a nickel crystallite at the end of the protruding "carbon fiber."

Since the heats of reaction for Reactions (1), (2), and (3) vary widely, it can be misleading to compare carbon-forming tendencies for these reactions by tabulating °F deviation from the temperature at which Reactions (1), (2), and (3) would be at equilibrium with respect to the actual product gas. A 200°F approach for Reaction (1) would be much farther away from equilibrium than the same 200°F approach for Reaction (2).

However, since all three reactions are pressure sensitive to the same degree and the water-gas shift reaction [(1) and (3) combination] is pressure insensitive, one can tabulate for any conditions the calculated pressure increase above which bulk carbon formation is suppressed by Reaction (2) and the pressure below which bulk carbon formation is suppressed by Reactions (1) and (3). If the water-gas shift reaction is in thermodynamic equilibrium, obviously the calculated pressure required to equilibrate Reactions (1) and (3) with respect to carbon formation will be identical. Let us call these dimensionless pressures P_1, P_2, and P_3 for Reactions (1), (2), and (3), respectively. Their definitions are:

$$P_1 = \frac{P_{CO_2} a_c}{P^2_{CO} K_1} = \frac{S(R + 1 + 2f)a_c}{f(1 - S)^2 P_T K_1}$$

$$P_2 = \frac{P_{CH_4} K_2}{P^2_{H_2} a_c} = \frac{(1 - f)(R + 1 + 2f)K_2}{f^2(3 + S)^2 P_T a_c}$$

$$P_3 = \frac{P_{H_2O} a_c}{P_{CO} P_{H_2} K_3} = \frac{(R - f - fS)\ (R + 1 + 2f) a_c}{f^2 (1 - S)\ (3 + S) P_T K_3}$$

where K_1, K_2, and K_3 are the thermodynamic equilibrium constants for Reactions (1), (2), and (3), respectively. Incorporating the mass balance equations:

$CO_2 = fS$, moles/mole CH_4 fed

$H_2 = 3f + CO = f(3 + S)$

$CH_4 = 1 - f$

$H_2O = R - f - fS$

$CO = f - fS$

Total moles $= R + 1 + 2f$

the ratios P_2/P_1 and P_2/P_3 then become simply:

$$\frac{P_2}{P_1} = \frac{K_1 K_2 (1 - f)\ (1 - S)^2}{fS(3 + S)^2 a_c^2} = \frac{K_1 K_2 P_{CH_4} P_{CO}^2}{P_{H_2}^2 P_{CO_2} a_c^2}$$

$$\frac{P_2}{P_3} = \frac{K_2 K_3 (1 - f)\ (1 - S)}{(3 + S)\ (R - f - fS) a_c^2} = \frac{K_2 K_3 P_{CH_4} P_{CO}}{P_{H_2} P_{H_2O} a_c^2}$$

where, in addition to K_1 (atm^{-1}), K_2 (atm), and K_3 (atm^{-1}), P_T is total pressure, f is the fractional methane conversion, R is the feed ratio of steam-to-methane (moles/mole), S is the selectivity of converted methane to carbon dioxide, $[S = CO_2/(CO + CO_2)]$, and a_C (= 1) is the activity of carbon.

These ratios then define the "net thermodynamic carbon forming tendency," and if the numerical value is equal to or greater than 1.0, bulk carbon will inevitably form, but may kinetically form at ratios less than 1.0. This exercise can be performed for partial conversions

Downloaded by [Tulane University] at 09:26 05 October 2013

of methane, assuming the water-gas shift is in equilibrium ($P_2/P_1 = P_2/P_3$) or by using actual partial methane conversion data where the results show that the water-gas shift is not equilibrated ($P_2P_1 > P_2P_3$), and the Boudouard Reaction (1) can cause serious carbon problems even when operation is above the minimum thermodynamically operable steam-to-carbon ratio.

These calculations show only the net thermodynamic driving force to form or remove bulk carbon; in actuality, relative kinetics may overrule. If, however, the approach to the water-gas shift is known as a function of operating variables, these calculations can be very useful in estimating conditions where carbon could lead to inoperability even when, at complete equilibration, the system would be operable. Certainly carbon will be formed, especially at the inlet of the reactor, if methane decomposition via Reaction (2) is kinetically faster than all the other reactions.

A theoretical example of this type of behavior is given in Fig. 6, where the pressure required to equilibrate Reactions (1), (2), and (3) with respect to bulk carbon formation (P_1P_T, P_2P_T, P_3P_T) is plotted against methane conversion, with, in this case, the water-gas shift assumed always to be in equilibrium ($P_1 = P_3$). The methane decomposition, Reaction (2), considered alone, which is the worst possible assumption, is carbon forming from 0 to 65% methane conversion at 1 atm total pressure. Likewise, it is carbon-forming from 0 to 23% conversion at 10 atm and from 0 to 7% at 100 atm. These conversions (65, 23, and 7) are approximately the equilibrium methane conversions permitted for the overall steam reforming reactions (Reactions A and B), so that methane decomposition alone is thermodynamically favorable over the entire range of permissible conversions. The Boudouard Reaction (1) and Reaction (3), since the water-gas shift has been assumed equilibrated, are carbon removing until 58% methane conversion has been achieved at atmospheric pressure, 27% at 10 atm, or 13% at 100 atm. The balance, then, or the net thermodynamic driving force to produce carbon ($P_2/P_1 = P_2/P_3$) at any pressure, is shown in the lower section of Fig. 6. It is quite apparent that at 1180°F, an inlet steam-to-carbon ratio of 1.2, and the water-gas shift at equilibrium, there is no net thermodynamic tendency to form carbon until 40% methane conversion has been achieved. Beyond 40% conversion, and up to 90%, a carbon-forming region exists, and carbon will inevitably form when the system is allowed to come to equilibrium within these conversion levels. At these conditions, operation above 3.2 atm (less than 40% conversion) is necessary in order to thermodynamically (although perhaps not kinetically) prevent bulk carbon formation.

Downloaded by [Tulane University] at 09:26 05 October 2013

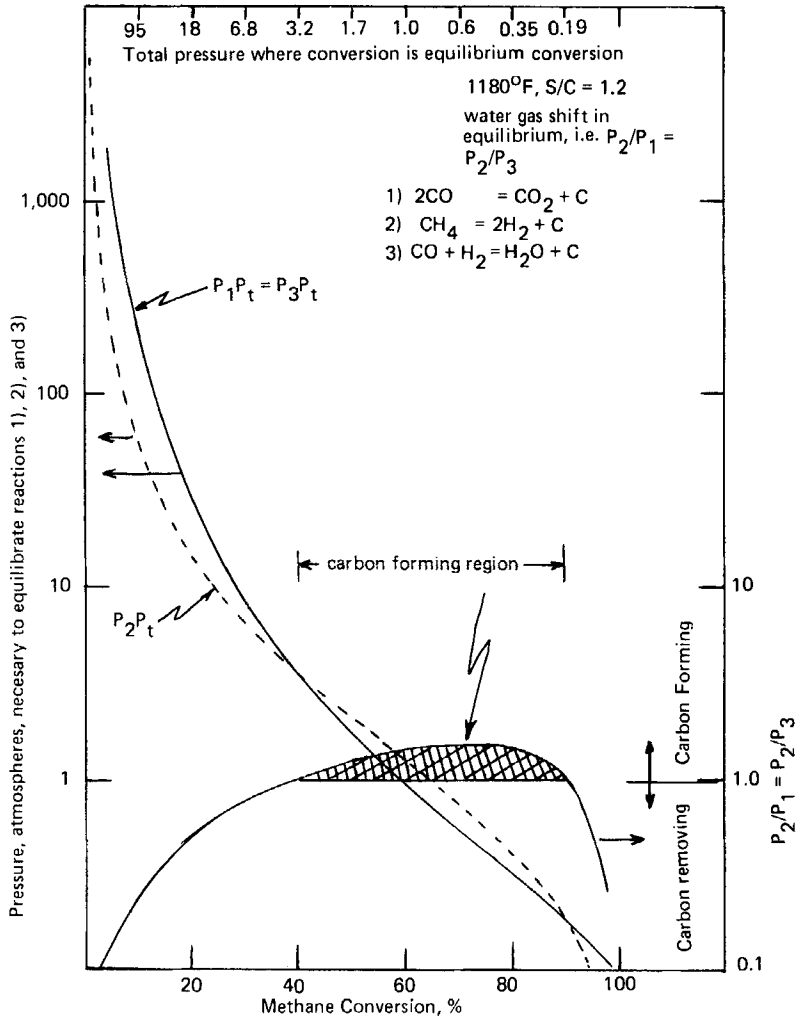


FIG. 6. Carbon-forming tendency in methane-steam reforming.

An example of applying this "carbon-forming tendency" technique to steam reforming/water-gas shift data, such as that of Akers and Camp, where the water-gas shift reaction is not equilibrated, is shown in Fig. 7. The net carbon forming tendencies, P_2/P_1 (carbon

Downloaded by [Tulane University] at 09:26 05 October 2013

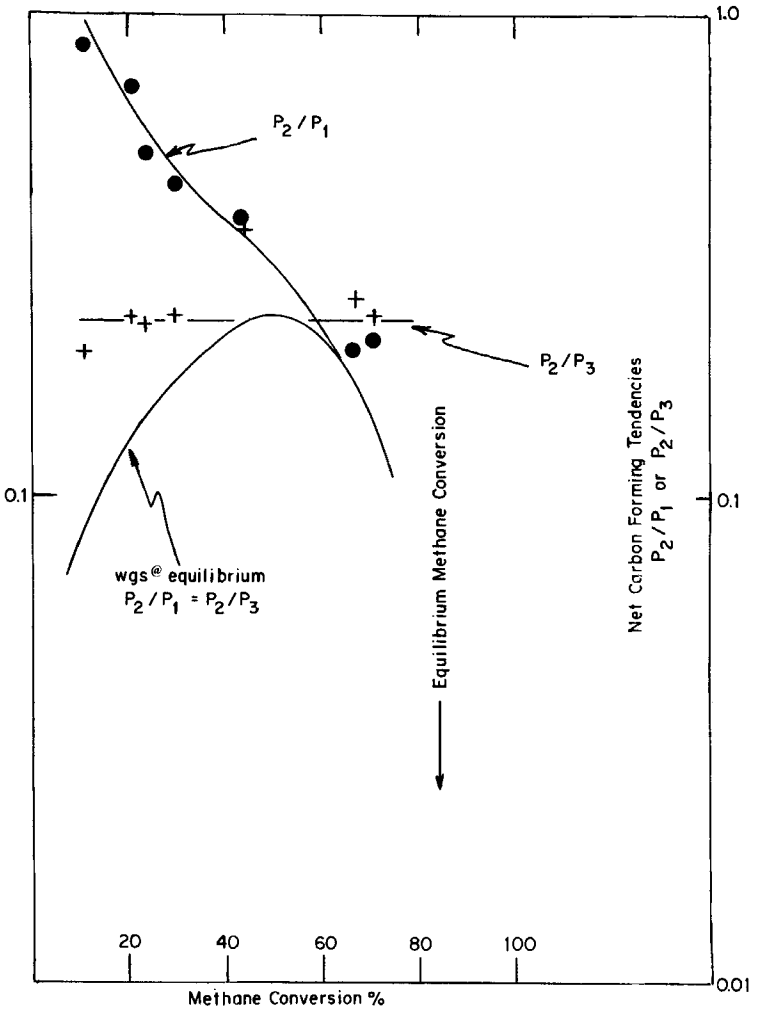


FIG. 7. Carbon-forming tendency vs methane conversion.

via methane and CO relative to carbon removal by CO_2 and H_2) and P_2/P_3 (carbon via methane and CO relative to carbon removal by steam and H_2) at 1180° F and S/C = 2.4 ± 0.3 are plotted as a function of methane conversion. For reference, the case when the

Downloaded by [Tulane University] at 09:26 05 October 2013

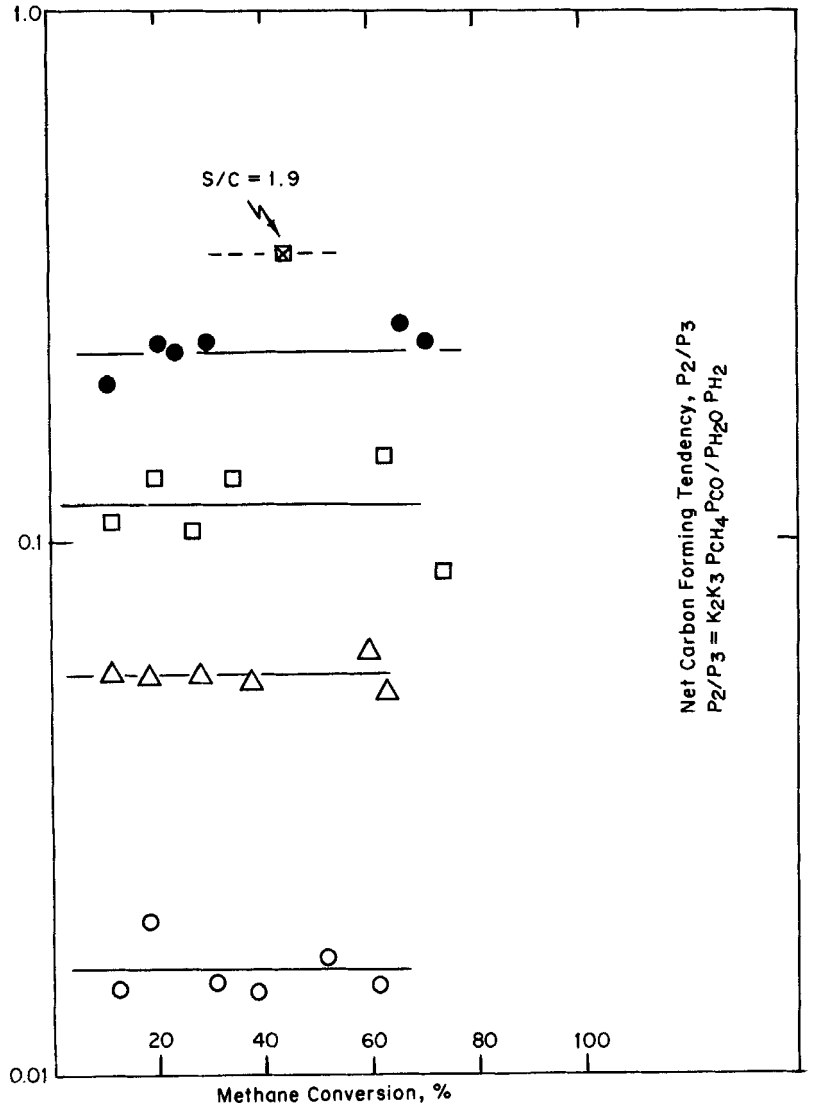


FIG. 8. Carbon-forming tendency: Methane or CO decomposition to carbon relative to carbon removal by steam or H_2, 1180°F and atmospheric pressure. After Ref. 35. See Fig. 3 legend.

Downloaded by [Tulane University] at 09:26 05 October 2013

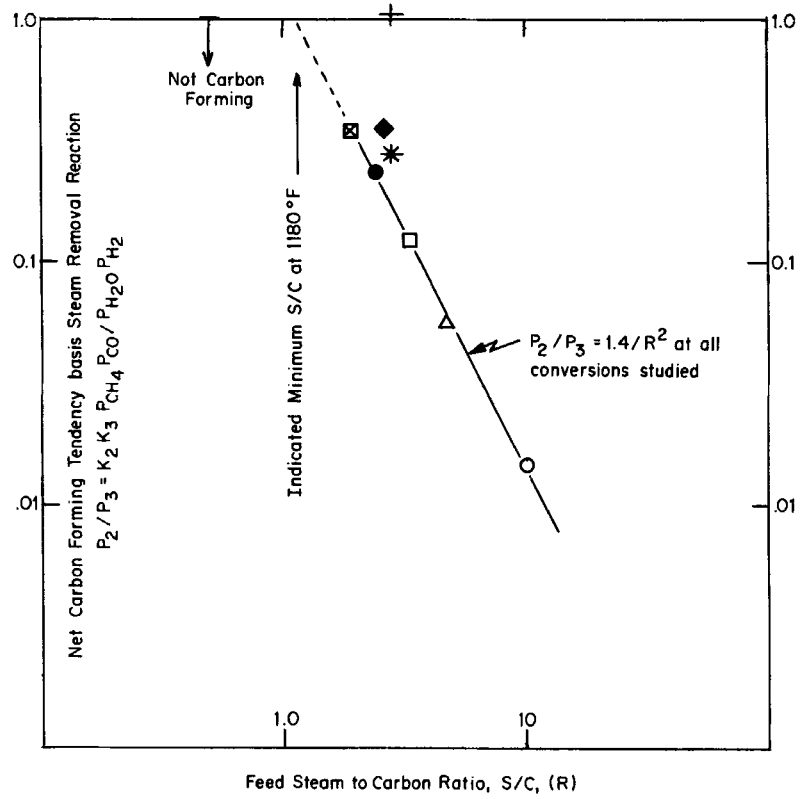


FIG. 9. Carbon-forming tendency in methane-steam reforming vs steam-to-carbon ratio, 1180°F, atmospheric pressure. After Akers and Camp [35]. See Fig. 4 legend.

water-gas shift reaction is equilibrated ($P_2/P_1 = P_2/P_3$) is also plotted. For all S/C ratios (9.9, 4.7, 3.3, and 2.4), P_2/P_3 was essentially constant, as in the example of Fig. 7, over the entire range of methane conversions studied (10 to 70% conversion) and varied inversely with the square of the S/C ratio. All S/C data are shown in Fig. 8, and a summary graph of the P_2/P_3 carbon-forming tendency is displayed in Fig. 9. Extrapolation of this line to the carbon-forming region ($P_2/P_3 = 1$) indicates a minimum steam requirement for steam-reforming methane at 1180°F to be about 1.2 moles/mole.

Downloaded by [Tulane University] at 09:26 05 October 2013

Three lower temperature data points indicate an increasing danger of forming carbon.

Assessment of the P_2/P_1 carbon-forming tendency (carbon removal by CO_2 and H_2) in a similar manner again shows that the carbon-forming tendency declines by the square of the S/C ratio, and also declines rapidly with increasing methane conversion. The expression developed for 1180° F operation is $P_2/P_1 = 10/e^{4.2fR^2}$, where f is fractional methane conversion and R is inlet steam-to-carbon ratio, moles per carbon atom. If carbon removal by CO_2 and H_2 is the operable limitation, extrapolation of this expression indicates a minimum steam-to-carbon ratio of 3.2, which is certainly high and also indicates coking at the reactor inlet, which is true only when the inlet temperature is high in methane-steam reforming. Therefore, P_2/P_3, including a lag in the water-gas shift reaction, appears to be the best measure of the potential for forming carbon.

V. KINETICS OF CATALYZED METHANE-STEAM REFORMING

Following Akers and Camp's work on the catalyzed steam-methane reforming reaction, which presumably was in the pore diffusion region (E_{act} = 15,800 Btu/lb mole), Bodrov et al. [39] studied the reaction of methane and steam using nickel foil in order to eliminate the effects of pore diffusion in the catalyst particles. A differential reactor was used and the temperature range of 1380 to 1650°F was covered at atmospheric pressure and S/C ratios of 1 to 20. The apparent activation energy for steam-reforming methane was determined to be 56,000 Btu/lb mole, and they suggested the rate-controlling step to be the decomposition of methane, forming chemisorbed CH_2 radicals. Bodrov et al. concluded that the intrinsic reaction rate is first-order in methane and the ratio of steam-to-hydrogen and also carbon monoxide pressure can influence the kinetic rate, especially at low temperature. This suggests an approach to an equilibrium expression, but the data were not tabulated so it cannot be tested.

In his Ph.D. thesis, Likins [40] studied the effect of pressure on the steam reforming of methane, using 3/16 X 3/16 in. cylinders of Girdler G-56B catalyst at 1180° F and a S/C ratio of 3.0. The total feed rate was kept essentially constant at 20 to 24 g moles per hour, and the catalyst charge was varied between 9 and 50 g. The effects of contact time and pressure from atmospheric to 18 atm were studied. Essentially all data were obtained beyond a 40% approach to

Downloaded by [Tulane University] at 09:26 05 October 2013

TABLE 1

Effect of Total Pressure on the Methane-Steam Reaction (after Likins [40])

Total pressure, atm	Initial methane conversion rate,[a] g mole/hr/g cat	Initial methane conversion rate,[a] g mole/hr/g cat/atm CH_4
1.0	0.1877	0.75
4.4	0.1934	0.176
7.8	0.1896	0.0974
11.2	0.1693	0.0605
14.6	0.1860	0.0510
18.0	0.1696	0.0377

[a]Polynomial fit technique.

Downloaded by [Tulane University] at 09:26 05 October 2013

methane equilibrium conversion (lowest absolute conversion was 22% at 18 atm, a 60% approach to equilibrium). The data were fitted by Likins to a third degree polynomial and initial reaction rates were extracted. Pore diffusion limitations were significant, as indicated by the pressure effect. The initial rates of reaction at various total pressures and the rate per atmosphere of methane pressure are given in Table 1.

Discounting several data points of Likins at a very high approach to steam-methane equilibrium conversion (as little as a 50°F drop in catalyst temperature can cause the calculated approach to equilibrium to change from 85 to 99+%), a good fit to simple reversible first-order kinetics (in the molecular pore diffusion region) is realized, ln [100/(100 - approach to equil)] = $k_{ov}t/f_e P_t$. The results are given in Table 2 in both units of sec^{-1} and g moles/hr/g cat/atm CH_4, and they show reasonable agreement with the polynomial fit of Likins.

As expected, in the molecular pore diffusion regime the apparent rate constant times the operating pressure is essentially constant ($k_{ov} \times P_t \cong 24$ ata/sec). The result of Likins' work at 1180°F, $k_{ov}P_t = 24$ ata/sec, using a catalyst of S/V ratio equal to 12.6 cm^{-1} compares favorably with the Akers and Camp result of 39 ata/sec

TABLE 2

Steam-Methane Kinetic Constants at Various Pressures
(after Likins [40])

Total pressure, ata	k_{ov}, sec^{-1} [a]	k_{ov}, g moles/g cat/hr/ata CH_4 [a]
1.0	19.1	0.46
4.4	5.5	0.13
7.8	5.5	0.13
11.2	2.1	0.051
14.6	1.7	0.041
18.0	1.5	0.036

[a] g moles CH_4 converted/g cat/hr/ata CH_4 X hr/3600 sec X d_p, g/cc X RT, cc ata/g mole = sec^{-1}, where d_p = catalyst pellet density (usually about 2 g/cc of pellet volume), R = 82 cc ata/g mole/°K, and T = °K.

with a catalyst of S/V of 18.9 cm^{-1}, and also with unpublished Kellogg Laboratory results using G-56B catalyst of S/V equal to 6.3 cm^{-1} (Table 3). The results are nearly linear in the S/V ratio. Kellogg's estimate of intrinsic activity of G-56B for steam-reforming methane at 1180°F and atmospheric pressure is 7800 sec^{-1}, based on the expression: $k_i^0 = 1.45 \times 10^{13} e^{-70,000/RT}$ sec^{-1}, determined from experimental measurements on finely divided catalyst up to temperatures approaching 1000°F.

A. Overall Kinetic Model for Steam Reforming and Water-Gas Shift Equilibration

When the kinetics are simple reversible first-order in methane in the intrinsic activity region (no effect of catalyst size), ln [100/(100 - % approach to equil)] = $k_i^0 t/f_e$ obtains. In the pore diffusion region where the rate is directly proportional to geometric catalyst surface area, the expression becomes:

$$\ln [100/(100 - \% \text{ approach to equil})] = \frac{k_{ov}^0 t}{P_t f_e}$$

Downloaded by [Tulane University] at 09:26 05 October 2013

TABLE 3

Effect of Catalyst Size at 1180°F

Observer	Catalyst S/V, cm^{-1}	Pressure, atm	$k_{ov}P_t/S/V$, $atm/sec/cm^{-1}$
Akers and Camp	18.9	1.0	2.1
Likins	12.6	1-18	1.9
Kellogg	6.3	1.0	1.6

Downloaded by [Tulane University] at 09:26 05 October 2013

where k^0_{ov} is the overall rate constant at atmospheric pressure and is, in its simplest form, a function of the surface-to-volume ratio of the catalyst pellet (S/V) and the intrinsic catalyst activity k^0_i. Total operating pressure P_t appears when the reaction takes place in the molecular pore diffusion region.

In the transition region between intrinsic kinetics and predominantly pore diffusion kinetics, i.e., $k_i \approx k_d$, the expression $k^0_{ov} = k^0_i \times k^0_d/(k^0_i + k^0_d)$ is effective in fitting the experimental data. The constants k^0_{ov}, k^0_i, and k^0_d are the overall specific rate constants for the reforming reaction, the intrinsic catalyst activity, and the activity in the pore diffusion region, respectively, at atmospheric pressure. The equation is derived via the simple electrical analog of additive resistances in series circuitry.

An example of this behavior over a very broad temperature range and showing the effect of catalyst particle size in the atmospheric pressure steam reforming of ethane using commercial Catalyst H is shown in Fig. 10. The data were obtained in the Kellogg Research and Development Laboratories. At atmospheric pressure and using full-size commercial rings, it is clear from the graph that the catalyst effectiveness factor (k^0_{ov}/k^0_i, where the superscript indicates the gauge pressure in atm) at 700° F is equal to 0.078, at 1000°F is about 0.0028, and at 1500°F has decreased to an estimated 0.00006. At 1500°F, with this low catalyst effectiveness of 0.006%, the indicated working depth of the commercial rings at atmospheric pressure calculates to be about 950 Å. Similar results are obtained with variously sized crushed catalysts, and they dramatically show the increase in overall rate constant as a function of catalyst size in the high-temperature regions. The activation energy for steam-reforming ethane in the low-temperature region or at higher temperatures using finely divided catalyst calculates to be about 56,000 Btu/lb mole, and shows a consistent decrease with increasing temperature. Unfortunately,

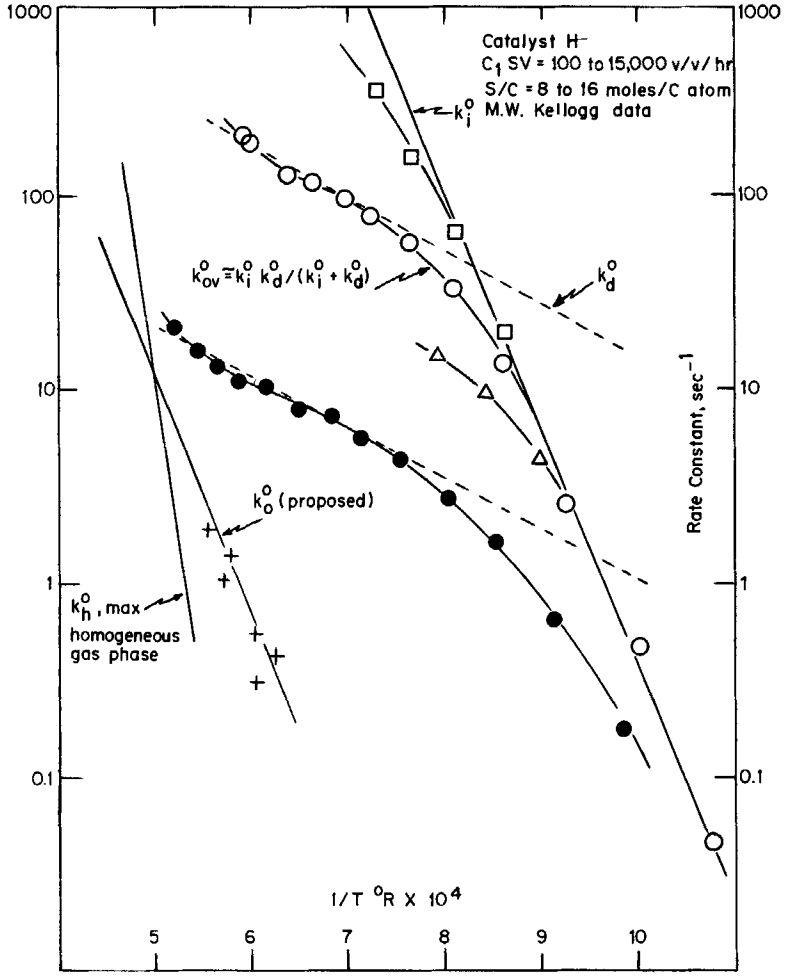


FIG. 10. Intrinsic and pore diffusion kinetics in ethane-steam reforming.

Size	S/V, cm^{-1}	Apparent Surface Roughness
□ 140/200 mesh	670	2
○ 12/20 mesh	48	2
△ 1/6 ring pieces	15	1.5
● 5/8 X 3/8 X 1/4" rings	6.3	1.0 (assumed)
+ Nickel tubing	28	1.0 (assumed)

Downloaded by [Tulane University] at 09:26 05 October 2013

the experiments were not conducted to still higher temperatures beyond 1450°F using rings, since the activation energy appears to be increasing in this high-temperature region.

This temperature region, however, was studied with a different, less active, commercial sized catalyst, Catalyst D. The results with Catalyst D rings (17 mm x 17 mm x 6 mm hole) and 12/20 mesh particles in the atmospheric pressure ethane-steam reaction are shown in Fig. 11. Here the increase in the activation energy in the high-temperature region is more apparent. The rate constant appears to be approaching a line which is described by a catalyst effectiveness factor of about 0.00044. This may relate to the steam-reforming reaction occurring on the "outside surface" of the ring, to a thermal homogeneous reaction showing its influence, or to both. Again the activation energy for the intrinsic steam reforming of ethane appears to be about 56,000 Btu/lb mole. If indeed it is the steam-reforming reaction on the catalyst "outside surface" in the high-temperature region, the reaction boundary depth can be simply estimated since the surface-to-volume ratio is known (4.8 cm^{-1}) and the effectiveness factor is estimated to be equal to 0.00044. The average reaction boundary depth is then about 9,000 Å.

Since the total surface area of Catalyst D is about 20 m^2/g, the average pore diameter is about 500 Å. Thus the "surface" of the ring behaves as if it is about 20 pore diameters deep, not an unreasonable figure in light of the highly porous nature of the catalyst, the probable binodal pore distribution, with the pore mouth walls acting as an external geometric surface. At any rate, the working reaction boundary depth appears to be very small in the high-temperature region, especially with catalysts which are intrinsically very active.

Now, returning to Fig. 10 and the data using Catalyst H, with an internal total surface area of about 50 m^2/g and an average pore diameter of 226 Å [41]. If the "outside surface" reaction is as depicted in Fig. 10, then the effectiveness factor in the high-temperature region appears to be approaching a value of 0.000025 and the average boundary depth would calculate to be 400 Å, which is about two average pore diameters. In line with this "outside surface" hypothesis is the data obtained on steam-reforming ethane at atmospheric pressure using solid forms of nickel where the effectiveness factor is equal to unity. These data are included in Figs. 10 and 11, and the rate constants obtained agree well with the proposed "outside surface" reaction on the large catalyst rings.

Supporting the effect of catalyst size, Allen et al. [41] have studied Catalyst H at 1180°F and atmospheric pressure in the methane-steam reforming reaction. They concluded that there was no effect of external bulk mass transfer of reactants to the catalyst

Downloaded by [Tulane University] at 09:26 05 October 2013

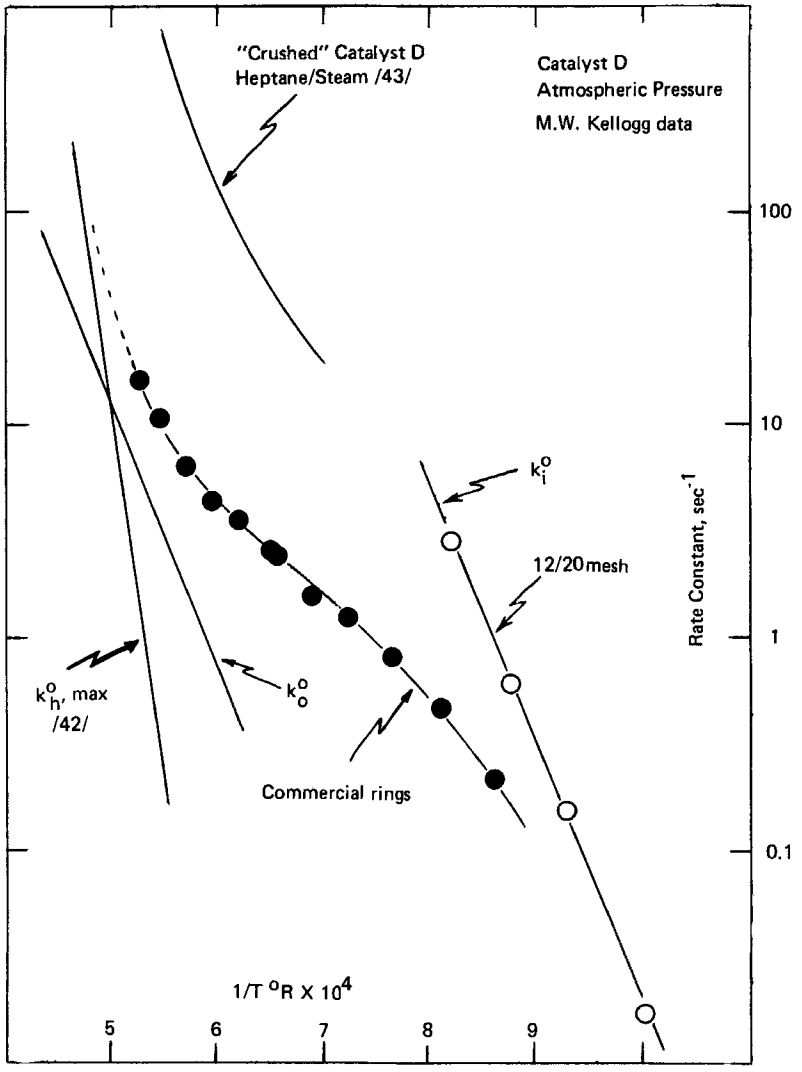


FIG. 11. Intrinsic and pore diffusion kinetics in ethane-steam reforming.

geometric surface. Using six different catalyst particle sizes ranging from 1/4 in. cylinders to 30/50 mesh crushed catalyst, they also concluded that pore diffusion plays a significant role in limiting the rate of

Downloaded by [Tulane University] at 09:26 05 October 2013

TABLE 4

Calculated Homogeneous Gas Phase Effect on the Rate in the Catalytic Steam-Reforming of Ethane (all rate constants in sec^{-1})

Temperature, °F	Intrinsic k_i^0	Pore diffusion, k_d^0	Outside surface, k_o^0	Gas phase, k_h^0	Thermal effect, $k_h^0/(k_d^0 + k_o^0)$
1,200	1,300	4.0	0.65	0.0035	0.00075
1,300	3,700	5.8	1.8	0.06	0.0079
1,400	8,500	7.5	4.2	0.6	0.051
1,500	18,000	10.0	9.2	5.5	0.29
1,600	35,000	13.0	18.0	35.0	1.1
1,700	65,000	15.5	34.0	180	3.6

Downloaded by [Tulane University] at 09:26 05 October 2013

the reaction. Their tabulated effectiveness factors (0.95 to 0.22, depending on size) are undoubtedly high since the reaction rate was still increasing linearly with reciprocal size even down to 30/50 mesh (0.45 mm average, S/V = 133 cm^{-1}).

Allen's initial rate data based on methane partial pressure (an apparent 2.6 steam-to-methane feed ratio was used) shows about 0.7 g moles CH_4 converted/g cat/hr/atm CH_4 for 1/8 in. cylinders, and 3.3 (equivalent to 140 sec^{-1}) for 30/50 mesh catalyst. Likins [40] obtained 0.75 g moles/hr/g/atm at atmospheric pressure in methane-steam reforming using Catalyst H (3/16 x 3/16 in. cylinders). Extrapolation of separate Kellogg results using Catalyst H would indicate that at 1180°F and atmospheric pressure, a catalyst particle size of 20 microns or smaller would have to be tested in steam-reforming methane in order to measure the intrinsic activity with little or no pore diffusion limitations.

To estimate the homogeneous gas phase thermal effects in the high-temperature region, Benson [42] gives the first-order rate constant (sec^{-1}) as a function of temperature for the carbon-carbon bond scission of ethane. The results of incorporating this estimate show that at about 1500 to 1600°F, the thermal rate may significantly increase the overall rate of the ethane-steam reaction. This calculated thermal effect in the steam-reforming ethane using Catalyst D rings (4.8 cm^{-1}) is shown in Table 4, and the thermal line, k_h^0, is indicated

in the Arrhenius plots of Figs. 10 and 11. With more active and/or smaller catalyst size, thermal effects will be relatively minor until a much higher temperature is attained. Also, methane feed will show proportionally less thermal effect than higher hydrocarbons. The apparent influence of thermal reaction on the steam reforming of heptane at atmospheric pressure using crushed Catalyst D is shown in Fig. 11 for comparison [43].

Another catalyst studied in the Kellogg Laboratories was commercial methane-reforming Catalyst G. The results of atmospheric pressure reforming tests using separate methane and ethane feeds are shown in Fig. 12. The results are typical for all catalysts tested when reforming is performed on both methane and ethane, and the rate constants are reproducible with the test temperature increasing and/or decreasing. In the intrinsic activity region there is a large difference in the rate constants between methane and ethane feed. Ethane-steam reforming is 10 to 20 times faster than methane. Typical of all the catalysts tested, the activation energy for steam-reforming methane is 70,000 Btu/lb mole and for ethane is 56,000 Btu/lb mole. In the pore diffusion region there is very little difference in the overall rate constant for either feedstock. Both show an activation energy of about 11,000 Btu/lb mole. Ethane again shows an upturn in the activation energy at about 1400 to 1500°F. At 1650°F, methane still has not shown a significant increase in the activation energy from its pore diffusion condition, since thermal effects would be less with methane and/or the reaction has not yet proceeded to the "outside surface" since the catalyst effectiveness factor in reforming methane is higher than for ethane. Because the activation energies are different, extrapolation to higher temperature would indicate that the reaction with methane feed should reach the "outside surface" at approximately 1700 to 1800°F, barring any strong resistance from bulk mass transfer of reactants to the catalyst external surface.

A summary of Kellogg data obtained in the strongly pore diffusion region for five different commercial catalysts ranging in intrinsic activity by a factor of about 300 is given in Fig. 13. Intrinsic activities in the high-temperature regions were extrapolated from lower temperature data as in Figs. 10, 11, and 12, although rate constants for Catalyst H as high as 350 sec^{-1}, using 140/200 mesh, were actually measured covering a range of four orders of magnitude for this catalyst and yielding first-order kinetics with an activation energy of 56,000 Btu/lb mole on ethane feed. Figure 13 shows not only the low power effect that increasing the intrinsic activity has on the actual measured constant in the pore diffusion regime, but indicates that alkali-promoted catalysts J, D, and E

Downloaded by [Tulane University] at 09:26 05 October 2013

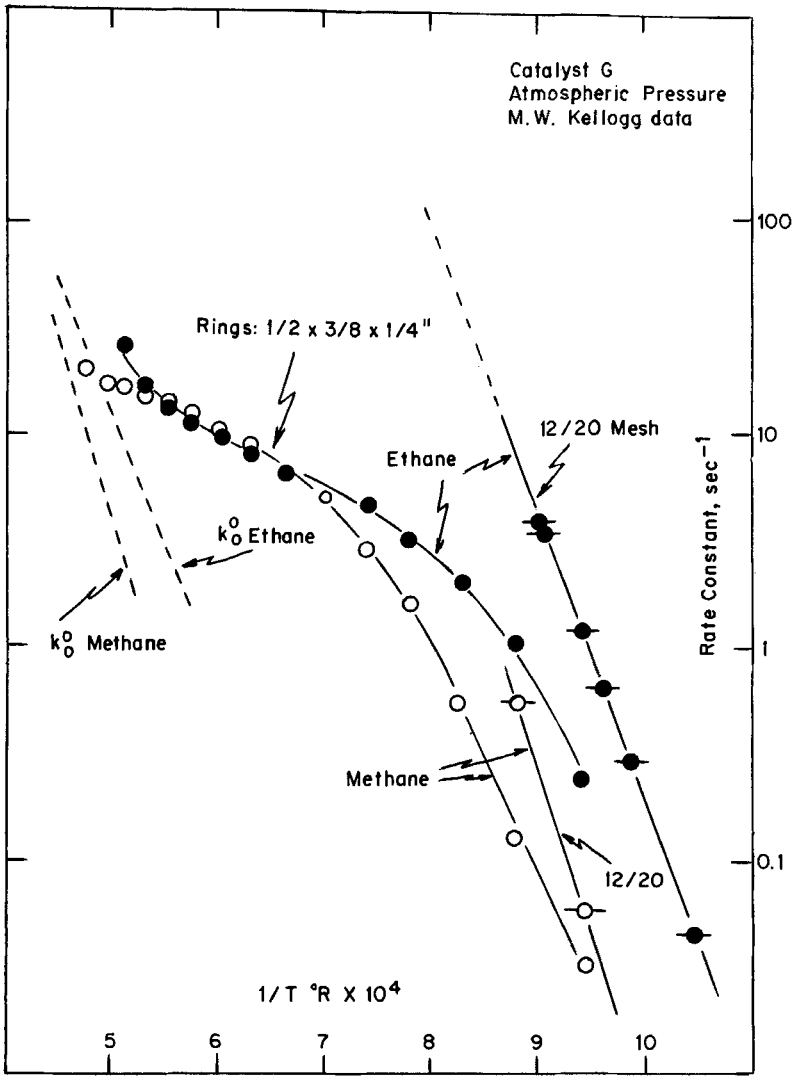


FIG. 12. Intrinsic and pore diffusion kinetics in ethane-steam reforming.

have, in the pore diffusion region, an activation energy one-third the activation energy for the intrinsic reaction and that nonalkalized catalysts H and G show about one-fifth the intrinsic activation energy.

Downloaded by [Tulane University] at 09:26 05 October 2013

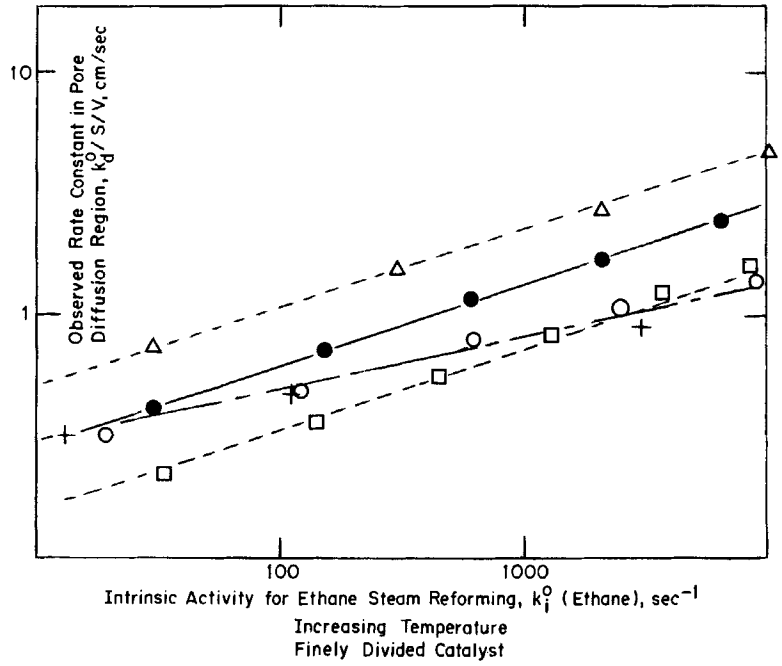


FIG. 13. Atmospheric pressure ethane-steam reforming. Rate constant in the pore diffusion region for large catalyst particles as a function of intrinsic catalyst activity at the same temperature.

Catalyst	k_i^o, 700° F	S/V, cm^{-1}	E_{act}, k_d^o, BTU/# mole
△ J, 1/8" cylinders	300 sec^{-1}	18.9	17,400
○ H, 5/8 X 3/8 X 1/4" rings	19 sec^{-1}	6.3	11,500
+ G, 1/2 X 3/8 X 3/16" rings	13 sec^{-1}	7.2	11,000
● E, 17mm X 17mm X 6mm rings	4.5 sec^{-1}	4.8	18,000
□ D, 17mm X 17mm X 6mm rings	0.9 sec^{-1}	4.8	19,800

This is not at all in agreement with what would be expected for a "simple reaction whose intrinsic kinetics can be expressed by a power law relationship" [44].

A general correlation of the observed catalyst activity times the operating pressure, $k_{ov}{}^{p}P_T$, atm/sec, in the pore diffusion region

Downloaded by [Tulane University] at 09:26 05 October 2013

as a function of the intrinsic activity k_i^0, sec^{-1}, for steam-reforming ethane at atmospheric pressure for an extreme range of catalyst activities (200,000-fold) studied in the Kellogg Laboratories is shown in Fig. 14. In general, for a given catalyst particle size, the observed activity in the pore diffusion region is roughly proportional to the one-third power of the intrinsic activity. Obviously, for low activity or active small particle size catalyst, the measured activity is equal to the intrinsic activity since this, by definition, is what is measured. A closer experimental examination of these widely varying catalysts would have to be undertaken to more accurately define observed activity as a function of intrinsic activity. As indicated previously, promoted catalysts apparently show a one-third power and nonpromoted catalysts a one-fifth power function.

Katsobashvili et al. [45] studied the steam reforming of methane using nickel catalysts in a fluid bed (0.3 to 0.4 mm size with 133 to 168 m^2/g total surface area). They showed that the catalyst is deactivated by steam oxidation (spinel formation) when more than 8 moles of steam per mole of methane is employed at 1200°F and atmospheric pressure. This type of deactivation is apparently "completely" reversible.

Catalysts were calcined at 1260°F prior to hydrogen reduction. Two-hour hydrogen reduction at atmospheric pressure and 1200°F or above was sufficient to activate the catalysts and reduce $NiO \cdot Al_2O_3$ spinel for these precipitated preparations. However, spinel-containing catalyst prepared by impregnation of alpha-alumina and calcined at a high temperature is not reduced by hydrogen below 1470°F [46].

The activity of several catalyst preparations was measured in order to determine the effect of nickel content on the activity. At a methane space velocity of 2400 cc/g cat/hr and an S/C ratio of 2.0, all catalysts containing from 10.2 to 54.4% nickel yielded essentially equilibrium conversion of methane from 1150 to 1500° F. Only the catalyst containing 3.4% nickel gave lower than equilibrium conversion at 1157 to 1200°F. This catalyst, again at a methane feed rate of 2400 cc/g/hr and an S/C of 2.0, yielded an 82% approach to steam-methane equilibrium at 1157°F and a 90% approach at 1200°F for indicated rate constants of 12.4 and 18.4 sec^{-1}, respectively, for a very approximate activation energy of 54,000 Btu/lb mole in this narrow temperature region.

In a series of experiments using the 10.2% nickel-on-alumina catalyst (0.4 mm) at 1200 to 1470° F, high feed rates were used (10,000 cc methane/g/hr at an S/C of 2.0) in an attempt to achieve less than equilibrium conversion. Even at the lowest temperature

Downloaded by [Tulane University] at 09:26 05 October 2013

Downloaded by [Tulane University] at 09:26 05 October 2013

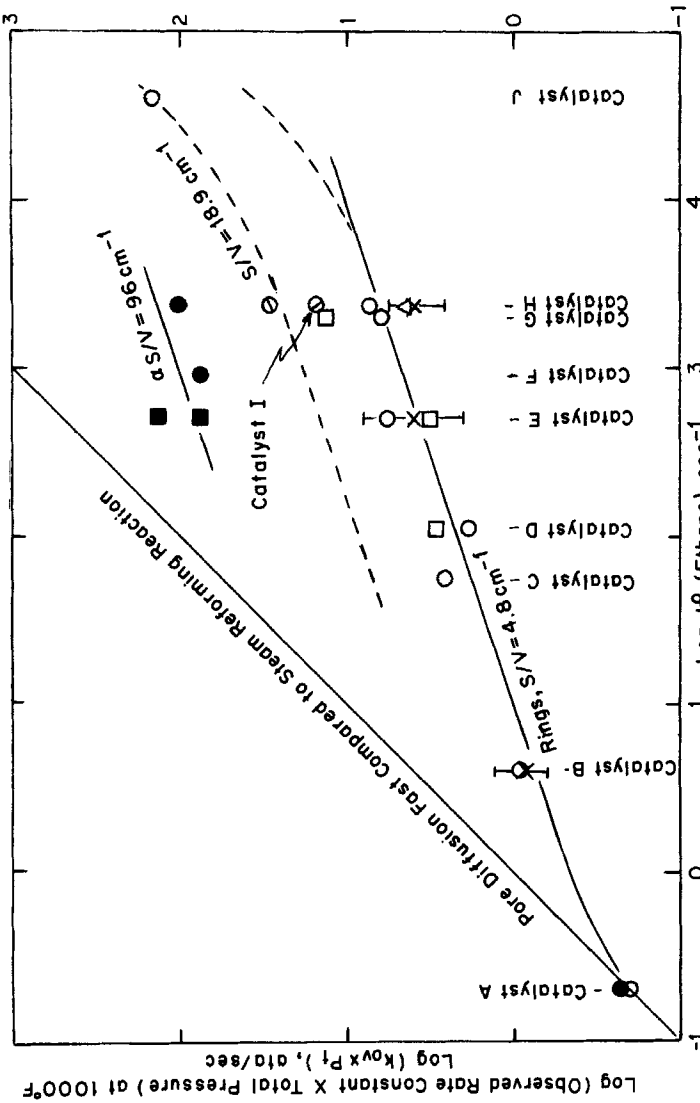


FIG. 14. Correlation of atmospheric intrinsic activity with high-pressure observed activity. M. W. Kellogg data. ○ Ethane, 1 ata (Rings & Cylinders); ● Ethane, 1 ata (12/20 mesh); ⊘ Ethane, 1 ata (1/6 ring pieces); × Methane, 1 to 40 ata (Rings); □ Naphtha, 32 ata (Rings); ■ Naphtha, 32 ata (12/20 mesh); △ Methane, 40 ata (Rings); ⊘ Ethane, 1 ata (1/4" cylinders).

of 1200°F, equilibrium conversion of methane was achieved. Under all conditions, both methane-steam and water-gas shifts were essentially equilibrated.

Following earlier studies on steam reforming, Bodrov and Apel'baum [47] reported on the carbon dioxide reforming of methane on nickel foil at atmospheric pressure and 1472 and 1652°F. They concluded that the rate of reaction of methane with carbon dioxide, at CO_2/CH_4 feed ratios equal to or greater than 1, is described by the same kinetic expression developed earlier [39]:

$$\text{Rate} = kP_{CH_4}/(1 + aP_{H_2O}/P_{H_2} + bP_{CO})$$

with an activation energy of 56,000 Btu/lb mole. They concluded that methane does not react with CO_2 but does react with steam, the hydrogen formed reacts rapidly with CO_2 via the water-gas shift reaction, and the steam is regained. They indicate that the initial reaction is one of methane decomposition to carbon-hydrogen deficient surface species wherein the initial hydrogen necessary for the water-gas shift reaction is formed. Analysis of Bodrov's data with respect to the approach to the water-gas shift reaction shows that steam is indeed approaching equilibration (from the steam-lean side) with respect to the water-gas shift reaction, carbon monoxide is approaching from the CO-lean side, and hydrogen is approaching from the H_2-rich side. These trends are shown in Fig. 15 where the approach to water-gas shift equilibration, especially for hydrogen (methane decomposition), is clearly indicated.

In order to investigate pore diffusion limitations, Bodrov et al. [48] continued their studies of the steam-methane reaction in a differential reactor at 1292 to 1652° F at atmospheric pressure using porous nickel catalysts of various sizes. In this temperature interval, where the equilibrium conversion of methane is greater than 90%, the kinetics were satisfactorily described by an irreversible first-order (in methane) reaction. Catalysis in the pore diffusion region was verified and consequently the rate of reaction of methane was directly proportional to the geometric surface area of the catalyst cubes. An activation energy of 35,000 Btu/lb mole was determined, well below the 56,000 Btu/lb mole which Bodrov previously had determined using nickel foil. Bodrov's data were collected over wide variations of operating variables (except pressure) including catalyst size and various experiments with N_2 and H_2 additions to the feed. Steam-to-carbon ratios were generally about 2 to 3 with a few runs at high S/C (12 to 16) and a few at low S/C (1.4 to 2). The approach

Downloaded by [Tulane University] at 09:26 05 October 2013

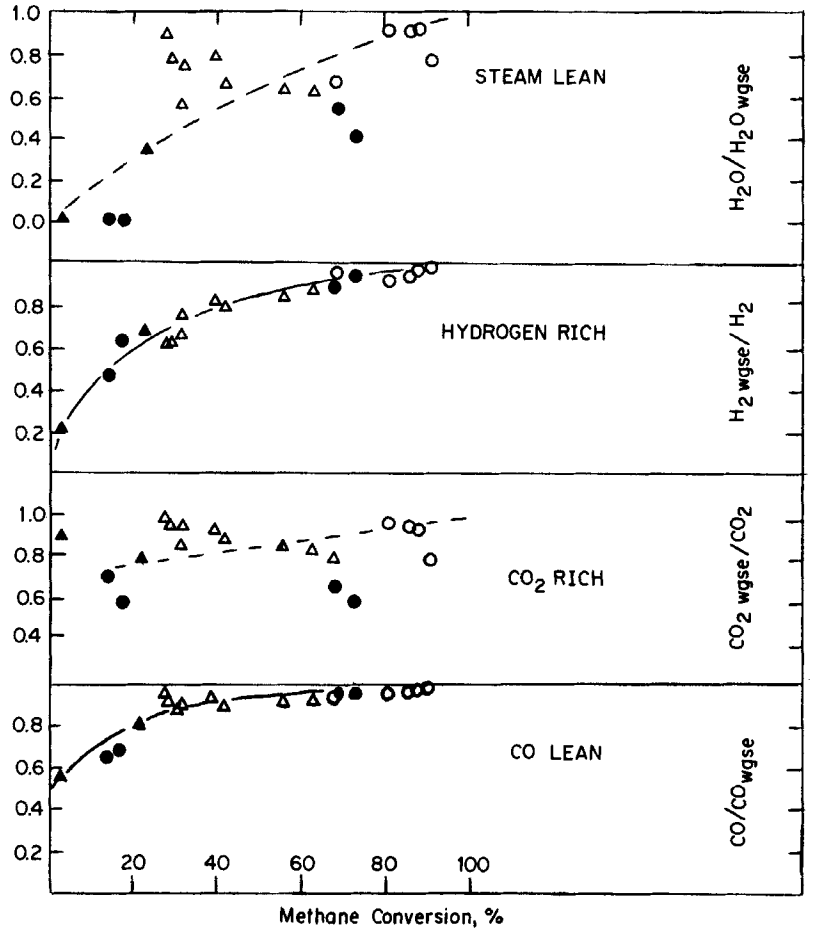


FIG. 15. Approach to water-gas shift for components in CO_2-methane reforming. After Bodrov [47]. Nickel foil, atmospheric pressure, O 1652°F, Δ 1472°F, Solid points: $CO_2/CH_4 < 1.0$. Open points: $CO_2/CH_4 > 1.0$.

to water-gas shift equilibrium in terms of the percent of CO that must be shifted to be in equilibrium is plotted against methane conversion in Fig. 16. This type of general correlation would be useful in establishing the gas composition at any methane conversion level,

Downloaded by [Tulane University] at 09:26 05 October 2013

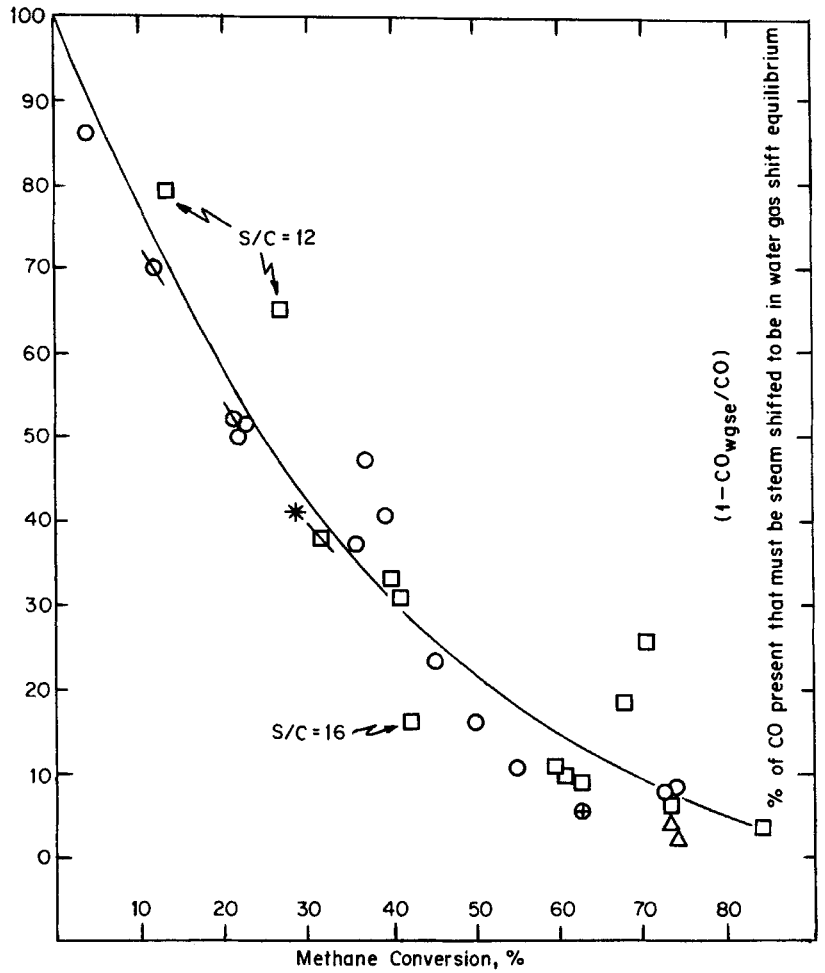


FIG. 16. Water-gas shift in methane-steam reforming. Porous nickel catalysts, atmospheric pressure. S/C = 2-3. After Bodrov [48]. O 1292°F; * 1382°F; □ 1472°F; ⊕ 1562°F; △ 1652°F; slashed = N_2 dilution.

irrespective of catalyst size, activity, temperature, and steam-to-carbon ratio. If applicable, only the methane conversion level need be known since the shift equilibrium constant and the mass balance equations can be used to calculate the complete gas composition

Downloaded by [Tulane University] at 09:26 05 October 2013

which is not in water-gas shift equilibrium. Consequently, the danger of forming carbon would be established. According to Fig. 16, at 20% methane conversion approximately 60% of the carbon monoxide present must be shifted in order to be in shift equilibrium. If the temperature at this methane conversion level is known, say 1480°F where $K_{wgs} = 1.0$, then the use of the mass balance equations yields

$$K_{wgs} = 1.0 = (CO_2 + x)\ (3f + CO_2 + x)/(R - f - CO_2 - x)\ (f - CO_2 - x)$$

and $x = 0.6(CO) = 0.6(f - CO_2)$ and, as previously, f = fractional methane conversion, R = molar steam-to-methane feed ratio, and x = moles of CO to be shifted per mole of methane feed. Solution of these equations then yields the moles of carbon dioxide (per mole of methane feed) present in the gas and consequently the entire gas composition can be calculated and the carbon-forming tendency established.

The experiments of Bodrov, wherein hydrogen was added to the feed, are shown in Fig. 17. In general, unless large amounts of H_2 are added to the methane feed (>3 moles/mole CH_4), there is little effect on the percent of carbon monoxide present that must be shifted to be in equilibrium.

When all the data previously discussed are presented on such a plot, a rather interesting "broad brush" correlation results. This is shown in Fig. 18. Although considerable scatter is apparent, these data represent methane-steam reforming over extreme ranges of pressure, temperature, steam-to-carbon ratios, catalyst size, stage of deactivation, and catalyst types, including nickel foil. Ross and Steel [54] data, which will be discussed later, are also presented in Fig. 18. They steam-reformed methane at the subatmospheric pressure of about 0.006 atm absolute.

Porous nickel catalysts were again used for steam-reforming methane at atmospheric pressure in continuing studies by Bodrov et al. [49] at lower temperatures (752 to 1112°F) than previously investigated in an attempt to determine the inhibitory effect of hydrogen. In a differential reactor, catalyst particle sizes from 0.4 to 5.3 mm were used at steam-to-methane ratio of 2 moles/mole. With catalyst particle size equal to or less than 0.4 mm, the reaction apparently proceeded in the intrinsic activity region with an activation energy of 65,000 Btu/lb mole. From the undiluted and N_2 and He diluted experiments (up to 4 moles of inert per mole of $CH_4 + H_2O$), Bodrov concluded that the reaction is first-order in methane and inhibited by hydrogen at the low temperatures. In the 752 to 932°F interval, Bodrov suggests rate = kP_{CH_4}/P_{H_2}, and at 1112°F, rate = $kP_{CH_4}/P_{H_2}^{0.5}$. At 1292°F and higher, where

Downloaded by [Tulane University] at 09:26 05 October 2013

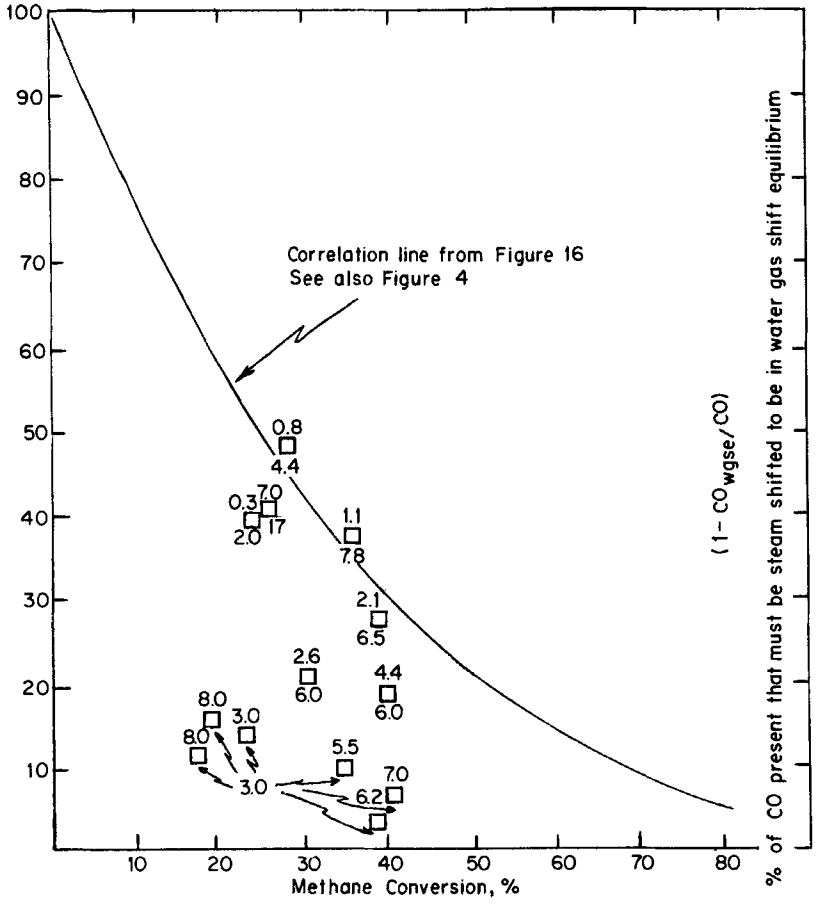


FIG. 17. Effect of hydrogen addition on the approach to water-gas shift equilibrium. Methane-steam reforming. After Bodrov [48]. Porous nickel catalysts, atmospheric pressure, 1472°F. Upper numbers: H_2/CH_4 feed ratio; lower numbers: S/C feed ratio.

hydrogen inhibition completely disappears, the expression would be straight "irreversible" first-order in methane (at 1292°F the equilibrium conversion of methane is approximately 95%).

Natural gas-steam reforming was studied by Atroshchenko et al. [50] at pressures of 2 and 6 atm using an integral reactor in the temperature range of 1112 to 1380° F at a steam-to-"gas" ratio of 2.0.

Downloaded by [Tulane University] at 09:26 05 October 2013

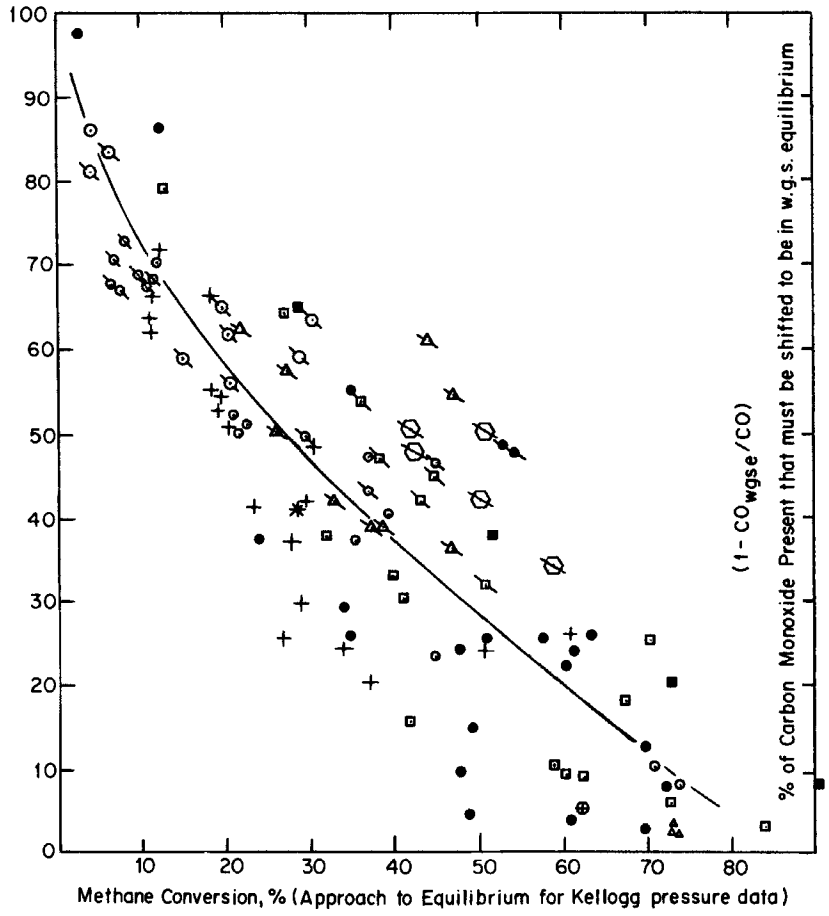


FIG. 18. General correlation of approach to water-gas shift as a function of methane conversion. ○ 1292°F, Bodrov [48]; * 1382°F, Bodrov [48]; □ 1472°F, Bodrov [48]; ⊕ 1562°F, Bodrov [48]; △ 1652°F, Bodrov [48]; + 1180, 943, 842° F, Akers and Camp [35]; ● 1470°F, Bodrov [39]; ■ 1112° F, Ross and Steel [57]; ⊘ 1000 - 1450°F, Rings, 1.0 Ata*; ⧋ 1000 - 1450°F, Rings, 6.7 Ata*; ⧈ 1000-1450°F, Rings, 20.2 Ata*; ⧂ 1000 - 1450° F, Rings, 40.5 Ata*; ◙ 1000 - 1450° F, Rings, 53.8 Ata*; *M. W. Kellogg data, Catalyst H.

Downloaded by [Tulane University] at 09:26 05 October 2013

The nickel catalyst used was 3 to 4 mm in size and had a "specific" surface of 5 m^2/g. Operating always at rather high approach to equilibrium methane conversions (>75% approach), Atroshchenko suggests the expression

$$-dP_{CH_4}/dt = kP_{CH_4}P_{H_2}^{-1}(1 - P_{CO}P_{H_2}^3/KpP_{CH_4}P_{H_2O})$$

for calculating the reaction rate at a close approach to equilibrium. The apparent activation energy was determined to be about 40,000 Btu/lb mole. Pore diffusion limitations were not investigated. Perhaps the first-order inverse hydrogen pressure term is, in reality, an inverse total pressure term due to molecular pore diffusion, since the authors operated at constant S/C ratio. Also, the transition region between intrinsic kinetics and strongly pore diffusionally controlled kinetics is indicated, with E_{act} = 40,000 Btu/lb mole.

Khomenko et al. [51], using a differential reactor at atmospheric pressure and 986°F with nickel foil as the catalyst, studied the methane-steam reaction and also the reverse reaction of methanation. Using the general expression,

rate (observed) = rate (forward) - rate (reverse)

for steam reforming,

$$R_f = R_{obs}/1 - (P_{CO}P_{H_2}^3/K_{eq}P_{CH_4}P_{H_2O})^{1/v}$$

and likewise for methanation we have,

$$R_f = R_{obs}/1 - (K_{eq}P_{CH_4}P_{H_2O}/P_{CO}P_{H_2}^3)^{1/v}$$

where v is the stoichiometric number of the reaction, and observed rates are rates of methane disappearance or appearance depending on the side from which equilibrium was approached. Equilibrium was approached from both sides at partial pressures of methane, steam, carbon monoxide, and hydrogen such that the denominator corrections in the above expressions were small if the stoichiometric number was equal to 1.0, i.e.,

$$R_f = R_{obs}/1 - 0.14, \quad \text{for steam reforming}$$

Downloaded by [Tulane University] at 09:26 05 October 2013

and

$$R_f = R_{obs}/1 - 0.15, \quad \text{for methanation}$$

If the stoichiometric number was equal to 2 or 3, considerable correction would result, and at various approaches to equilibrium an inconsistency between forward rate and observed rate would result. On the basis of the experimental results, the authors concluded that the stoichiometric number is equal to unity. At approximately constant partial pressures of steam (P_{H_2O} = 0.2 ata) and hydrogen (P_{H_2} = 0.2 ata), they found that the rate of methane disappearance was directly proportional to the methane partial pressure from 0.01 to 0.1 ata and independent of the CO and CO_2 content of the mixture. Also, in methanation, the rate of methane formation was directly proportional to the carbon monoxide partial pressure from 0.01 to 0.1 ata and independent of methane partial pressure. At 986°F the measured rates were 0.39 g moles CH_4 reformed/hr/m^2 of Ni/ata CH_4 and 0.17 g moles CH_4 produced/hr/m^2 of Ni/ata CO for steam reforming and methanation, respectively. The authors state that this value for steam reforming is in good agreement with that obtained using finely divided porous nickel catalysts where the nickel surface area was known.

At 986°F the authors conclude that the rate of steam reforming methane is

$$R_f = kP_{CH_4} f(H_2, H_2O)$$

and the rate of methane formation via the methanation reaction is

$$R_f = k'P_{CO} f(H_2, H_2O)$$

and indicate that the function of $f(H_2, H_2O)$, based on the unreported data [53], is of a rather complex form.

In the region where both approaches to equilibrium for methane-steam reforming and carbon monoxide methanation were the same (ΔT = 100°F), the approach to water-gas shift equilibrium was approximately 50° F. At an 180° F approach to steam-methane equilibrium (methanation), $^{\Delta}T_{wgs}$ was equal to -100°F (CO rich), and at a 73°F steam-methane approach (methane-steam reforming), $^{\Delta}T_{wgs}$ was equal to -4°F (slightly CO rich), indicating that $^{\Delta}T_{wgs}$ is approximately one-half of the ΔT (steam reforming and methanation).

High temperature (1342 to 1882° F), low steam-to-carbon ratio (1.2-1.5), steam reforming of natural gas was studied by Nekrich

Downloaded by [Tulane University] at 09:26 05 October 2013

[52] who was interested in the production of metal ore reducing gas. Atmospheric pressure and crushed commercial catalyst of 1.0 to 1.5 mm size was employed in an integral reactor. Methane conversion levels from 20 to 90% were obtained at contact times of 0.01 to 0.1 sec. There were no apparent problems associated with carbon formation, continuous operating times of about 200 hours being achieved. The author states that accidental contact of water vapor or air with the catalyst, in the absence of natural gas, apparently led at high temperatures to formation of nickel spinel, $NiO \cdot Al_2O_3$, which is resistant to regeneration. For kinetics, the author concluded that a reliable empirical expression for methane-steam reforming at atmospheric pressure, low S/C ratio, and 1342 to 1880°F operation is

$$-dCH_4/dt = k(CH_4 + H_2O)/(2H_2 + CO)$$

showing first-order kinetics in methane and steam and first-order inhibition terms in hydrogen and carbon monoxide. Nekrich states that the same results were obtained using different catalyst size, indicating kinetic control of the reaction. An apparent activation energy of 32,000 Btu/lb mole for methane-steam reforming was determined.

Khomenko et al. [53] continued studies on methane-steam reforming and methanation in a further attempt to elucidate the reaction kinetics and quantitatively evaluate the form of the steam and hydrogen inhibition terms in the rate expression previously suggested [51]. Atmospheric pressure and nickel foil catalyst were used in order to determine the intrinsic kinetics in the absence of pore diffusion. A differential reactor was again used and temperatures of 878, 986, 1112, and 1292°F were studied. A six-step reaction scheme was proposed and the kinetics of steam reforming derived in which the forward rate is directly proportional to both steam and methane pressures (with standard approach to equilibrium term) and inversely proportional to steam and hydrogen polynomial terms. Indicated activation energy at low methane conversion was 60,900 Btu/lb mole.

Using finely divided, presumably very active nickel catalyst (coprecipitated 75% Ni/1% K + Na/alumina, 250/355 micron particle size, 37 m^2/g nickel area, and total surface area of 175 m^2/g), Ross and Steel [54] studied the steam reforming of methane at very low subatmospheric pressure (0.0025 to 0.013 ata) in a static apparatus. Operating in most cases at low S/C ratios (2 to as low as 0.2), conditions were such that there was always more than one

Downloaded by [Tulane University] at 09:26 05 October 2013

TABLE 5

Tendency toward Carbon in Very Low Pressure Methane-Steam Reforming (after Ross and Steel [54])

Methane conversion, %	P_2/P_1	P_2/P_3
28	158	2.8
52	50	2.3
73	18	2.0
91	5.8	1.2

Downloaded by [Tulane University] at 09:26 05 October 2013

surface nickel atom per reactant ($CH_4 + H_2O$) molecule in the system. At 1112°F unusually low methane conversion rates of about 0.5 g moles methane converted/g cat/hr/ata methane were obtained for this very finely divided catalyst. An activation energy of 52,000 Btu/lb mole was estimated, although there may have been some influence of Knudsen pore diffusion limitations at this low pressure. The authors concluded that the rate-determining step on a fully reduced catalyst under reducing conditions (low S/C ratio) is the rate of adsorption of methane and competition for the adsorption sites by water vapor. They suggested that the steam-methane reforming reaction is first-order in methane and inverse one-half power in steam. Gas composition data showed a smooth trend of closer approach to water-gas shift equilibrium with increasing methane conversion (see Fig. 18). At these very low absolute pressures, the data are substantially the same as the dramatically higher pressure data, further suggesting a general trend correlation of approach to water-gas shift equilibrium as a function of simply the methane conversion (approach).

At a typical condition listed by Ross and Steel [54] (S/C = 1.0, initial total pressure = 0.0062 ata), the net thermodynamic carbon-forming tendencies (P_2/P_1 and P_2/P_3) for all reactions involved (see Figs. 6 and 7) were considerably greater than unity for all methane conversion levels, indicating that carbon formation may have been a serious problem in this work. Data taken from Ross and Steel's report showing this unusually large carbon-forming tendency is given in Table 5.

B. Summary of Temperature Effect in Steam-Methane Reforming

See Table 6.

Downloaded by [Tulane University] at 09:26 05 October 2013

TABLE 6

Summary of Temperature Effect in Steam-Methane Reforming

Investigator	Catalyst	Temperature, °F	E_{act}, Btu/lb mole	Comments
Fujimoto [26]	Ni foil	1600	98,000	Bulk carbon removal?
Fujimoto [26]	Ni foil	1300-1650	78,000	Methane decomposition?
Fujimoto [26]	Ni/Al_2O_3	750-1100	9,500	Pore diffusion?
Lewis [30]	Cu/SiO_2	1400-1600	86,000	Finely divided catalyst
Arnold [34]	Ni/support	1300-1500	≥ 60,000	16/20 mesh catalyst
Arnold [34]	Ni/support	1400-1550	~ 26,000	7/16 X 7/16 in. catalyst
Akers [35]	Ni/kieselguhr	750-1180	15,800	1/8 in. catalyst
Bodrov [39, 47]	Ni foil	1380-1650	56,000	Back pressure terms
Likins [40]	Ni/support	1180	-	Total pressure effect
Kellogg (Fig. 12)	Ni/support	600-700	70,000	12/20 mesh catalyst
Kellogg (Fig. 12)	Ni/support	1100-1600	11,000	Commercial rings
Katsobashvili [45]	Ni/support	1160-1200	54,000	0.3/0.4 mm catalyst
Bodrov [48]	Ni/support	1290-1650	35,000	Catalyst size effect
Bodrov [49]	Ni/support	750-1110	65,000	Small catalyst low temperature
Atroshchenko [50]	Ni/support	1110-1380	40,000	3-4 mm catalyst
Nekrich [52]	Ni/support	1340-1880	32,000	1-1.5 mm catalyst
Khomenko [53]	Ni foil	880-1290	60,900	Back-pressure terms
Ross and Steel [54]	Ni/support	1100	52,000	Very fine catalyst

VI. CONCLUDING REMARKS

The literature has been reviewed on the subject of kinetics of methane-steam reforming involving integral and differential reactor data, porous nickel catalysts and nickel foil, and data over large ranges of temperature (500 to 1700°F), pressure (0.01 to 50 atm), and intrinsic catalyst activities (200,000-fold). Considerable heretofore unpublished Pullman Kellogg laboratory data correlations are presented. A simple reversible first-order kinetic expression for the steam-methane reaction appears to be applicable throughout the operable region of steam-to-carbon ratios. Internal pore diffusion limitation on the conversion rate, due to catalyst size and/or intrinsic catalyst activity and total operating pressure, is underlined. "s"-shaped Arrhenius plots (changing activation energy) are obtained when steam reforming is conducted over a temperature range sufficient to produce intrinsic kinetics (low temperature, inactive catalyst, or small catalyst size), pore diffusional limitations (large catalyst size or active catalyst), and "reaction on the outside surface" (very low catalyst effectiveness factor). Homogeneous gas-phase kinetics appear to contribute only at relatively high temperature (1400°F).

In steam reforming, the water-gas shift reaction departs from its equilibrium position (carbon monoxide rich), especially at low methane conversion level. A general correlation of approach to water-gas shift equilibration as a function of conversion level only is indicated.

Acknowledgment

The previously unpublished Pullman Kellogg laboratory results and data presented herein are a small segment of the data and experience accumulated by Pullman Kellogg through many years of continuous research and development activities related to reforming catalyst and reforming processes. The author wishes to thank Pullman Kellogg for their permission to use their data in this paper.

Downloaded by [Tulane University] at 09:26 05 October 2013

REFERENCES

[1] P. J. Byrne, R. J. Gehr, and R. T. Haslam, Ind. Eng. Chem., 24, 1129 (1932).

[2] N. R. Gard, Nitrogen, 39, 25 (1966).

[3] S. P. S. Andrew, Eastbourne Conference on Fuel Research and Development, Institute of Fuel, London, 1965.

[4] G. W. Bridger, N. R. Gard, and H. Watson, Chem. Eng. World, 5(6), 93-101, 126 (1970).
[5] J. R. Rostrup-Nielson, Steam Reforming Catalysts, Danish Technical Press, Copenhagen, 1975.
[6] L. F. Marek and D. A. Hahn, Catalytic Oxidation of Organic Compounds in the Vapor Phase, Chemical Catalog Co., 1932, p. 101.
[7] H. Davy, Philos. Trans. R. Soc. London, 107, 77 (1817).
[8] M. M. Tessie Du Motay and Marechal, Bull. Soc. Chim. Fr., 9, 334 (1868).
[9] L. Mond and C. Langer, German Patent 51,572 (1889); Chem. Zentralbl., 2, 32 (1890).
[10] Kirk-Othmer, Encyclopedia of Chemical Technology, 2nd ed., Vol. 10, 1966, p. 353.
[11] See Ref. 6, p. 102.
[12] L. Mond and C. Langer, British Patent 12,608 (1888).
[13] C. Dieffenbach and W. Moldenhauser, German Patent 229,406 (1909).
[14] A. Mittasch and C. Schneider, U.S. Patent 1,128,804 (1915), to BASF.
[15] British Patent 267,535 (1927), to I. G. Farben.
[16] P. Sabatier, Catalysis in Organic Chemistry (translated by R. E. Reid), Van Nostrand, New York, 1922, Chap. VIII. Paragraphs 392-402; Chap. IX, Paragraphs 409-411, Paragraph 614; Chap. XXI, Paragraphs 911-912.
[17] See Ref. 6, Chap. IX.
[18] Lewis and Randall, Thermodynamics and Free Energy of Chemical Substances, McGraw-Hill, New York, 1923.
[19] B. Neumann and K. Jacob, Z. Elektrochem., 30, 557 (1924).
[20] R. N. Pease and P. R. Cheseboro, J. Am. Chem. Soc., 50, 1464 (1928).
[21] H. Pichler, Z. Angew. Chem., p. 472 (June 6, 1931).
[22] F. Fischer and H. Tropsch, Brennst.-Chem., 9, 39 (1928).
[23] C. O. Hawk, P. L. Golden, H. H. Storch, and A. C. Fieldner, Ind. Eng. Chem., 24, 23 (1932).
[24] W. A. Karzhavin, Ibid., 28, 1042 (1936).
[25] K. S. M. Bhatta and G. M. Dixon, Trans. Faraday Soc., 63, 2217 (1967).
[26] H. Fujimoto, J. Soc. Chem. Ind. Jpn., Suppl., 36, 147-148 (1933).
[27] P. L. Walker, Jr., F. Rusinko, Jr., and L. G. Austin, Adv. Catal., 11, 159-160 (1959).
[28] A. S. Gordon, Ind. Eng. Chem., 38, 718 (1946).

Downloaded by [Tulane University] at 09:26 05 October 2013

[29] R. E. Reitmeier, K. Atwood, H. A. Bennet, Jr., and H. M. Baugh, Ibid., 40, 620 (1948).
[30] W. K. Lewis, E. R. Gilliland, and W. A. Reed, Ibid., 41, 1227 (1949).
[31] V. N. Ipatieff, G. S. Monroe, and L. E. Fischer, Ibid., 42, 92 (1950).
[32] J. J. S. Sabastian and C. H. Riesz, Ibid., 43, 860 (1951).
[33] C. H. Riesz, H. A. Dirksen, and W. J. Pleticka, Pet. Eng., 24(13), C-17 (1952).
[34] M. R. Arnold, K. Atwood, H. M. Baugh, and H. D. Smyser, Ind. Eng. Chem., 44, 999 (1952).
[35] W. W. Akers and D. P. Camp, AIChE J., 1, 471 (1955).
[36] L. J. Petrovic and G. Thodos, Ind. Eng. Chem., Fundam., 7, 274 (1968); see also Ref. 44, p. 82.
[37] F. J. Dent, "49th Report of the Joint Committee," Trans. Inst. Gas Eng., 95, 604 (1945-46).
[38] G. W. Bridger and Chinchen, Catalyst Handbook, Springer, New York, 1970, Chap. 5, pp. 73-76.
[39] N. M. Bodrov, L. O. Apel'baum, and M. I. Temkin, Kinet. Catal., 5, 614 (1964).
[40] M. R. Likins, Jr., Ph.D. Thesis, University of Louisville, 1970.
[41] D. W. Allen, E. R. Gerhard, and M. R. Likins, Jr., Br. Chem. Eng. Process Technol., 17(7/8), 605 (1972).
[42] S. W. Benson, The Foundations of Chemical Kinetics, McGraw-Hill, New York, 1960, p. 354.
[43] G. W. Bridger, N. R. Gard, and H. Watson, Chem. Eng. World, 5(6), 93 (1970).
[44] C. N. Satterfield, Mass Transfer in Heterogeneous Catalysis, MIT Press, Massachusetts, 1970, p. 4.
[45] Ya. R. Katsobashvili, A. R. Brun-Tsekhovoi, V. V. Shchekin, and I. I. Sladkovskaya, Kinet. Catal., 2, 513 (1961).
[46] A. G. Leibush et al., Khim. Prom, 2, 16 (1961); see also Ref. 45.
[47] I. M. Bodrov and L. O. Apel'baum, Kinet. Catal., 8, 326 (1967).
[48] I. M. Bodrov, L. O. Apel'baum, and M. I. Temkin, Ibid., 8, 696 (1967).
[49] I. M. Bodrov, L. O. Apel'baum, and M. I. Temkin, Ibid., 9, 877 (1968).
[50] V. I. Atroshchenko, Sh. K. Raman, and G. L. Zvyagintsev, J. Appl. Chem. USSR (Engl.), 42(7), 1414 (1969).

Downloaded by [Tulane University] at 09:26 05 October 2013

[51] A. A. Khomenko, L. O. Apel'baum, F. S. Shub, and M. I. Temkin, Kinet. Catal., 11, 1233 (1970).
[52] E. M. Nekrich, J. Appl. Chem. USSR (Engl.), 43(2), 372 (1970).
[53] A. A. Khomenko, L. O. Apel'baum, F. S. Shub, Yu. S. Snagovskii, and M. I. Temkin, Kinet. Catal., 12, 367 (1971).
[54] J. R. H. Ross and M. C. F. Steel, J. Chem. Soc., Faraday Trans. 1, 1, 10 (1973).

Downloaded by [Tulane University] at 09:26 05 October 2013